

Alternating Transmission Protocol

Receiver-Dependent Allocation of Rapid and Recurrent Processing

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August 13, 2026

Preprint; not peer reviewed.

Abstract

A nervous system cannot allocate the same amount of return processing to every event. Some actions can proceed from recent evidence; some require earlier context, correction, or recurrent stabilization; and some uncertainties cannot be repaired by more computation. The scientific problem is therefore not merely whether recurrence exists, but how a receiver-dependent system allocates rapid and return influence when accuracy, calibration, action value, delay, computation, safety, and energetic cost can disagree. This paper develops that problem from the Alternating Transmission Protocol (ATP), named in a 2017 *Neural Lace Podcast* discussion and traced to receiver-dependent sense-action and neural-route sources from 2011–2012. The original Internet-transport analogy is translated into a biological and computational regime vector comprising forward influence, return influence, controlled-loss tolerance, order sensitivity, correction, persistence, receiver readiness, action coupling, energy, and latency. Later Phase Wave Differential (PWD) and Neural Array Projection Oscillation Tomography (NAPOT) formulations supply distinct receiver-relative and multi-array hypotheses; they are not synonyms for ATP. The vocabulary is introduced only after the problem, established mechanisms, and proposed operation have been explained. Four preregistered synthetic development studies progressively strengthened the test. Structured return content, receiver-state features, and conditional allocation affected outcomes, but the tested hysteretic controllers did not establish unique effectiveness. In the strongest nonlinear navigation study, the ATP candidate reached 0.9078 exact accuracy, 0.9149 task action score, and 0.7815 registered utility while using return processing on 19.69% of trials. It exceeded a trajectory expected-value gate in mean utility, but missed the complete conjunctive endpoint and was outperformed by the simpler direct final expected-value policy, which achieved 0.7957 utility with less computation. Every tie, null result, failed margin, adverse ablation, and unrealized workload condition is retained. The work also provides a finite machine-checked policy kernel, explicit nonidentification theorems, strong comparisons with predictive coding, adaptive computation, early exit, learned stopping, and recurrent processing, and a public robot-data preflight. The public sources contain real sequences and outcomes but do not join randomized route assignment, route-caused action change, executed action, and observed consequence in the same governed episode. A prospective physical action-consequence experiment is therefore specified and remains unrun. The present contribution is a source-recovered theory-and-methods framework, an executable and reproducible negative-result program, a bounded formal verification layer, and a discriminating experimental design—not evidence that brains implement the finite controller or that ATP already improves a physical robot.

Keywords: neural communication; recurrent processing; feedforward processing; adaptive computation; calibration; action consequence; thalamocortical loops; predictive coding; brain-computer interface; Self-Aware Networks; NAPOT; Phase Wave Differential

1. A familiar problem: when should a signal return?

Consider three ordinary situations. While walking, a few visual details can disappear behind an obstacle without causing the scene to collapse. When recognizing an object in poor light, however, the first sensory pass may be insufficient; context and later processing alter the result. When holding an intended movement through a delay, an activity pattern must survive after the original cue is gone. All three cases move information through a network, yet they do not place the same demand on that network.

The first case rewards continuity, speed, and tolerance of incomplete input. The second rewards comparison and correction. The third rewards persistence. A single instruction such as “send the signal” hides these differences. The brain must decide—locally and across interacting regions—whether a current relation can continue through a rapid forward transformation or whether it requires renewed influence from other levels, repeated exchange, state-dependent amplification, or a return path.

Established neuroscience already supplies many pieces of this problem. Predictive-coding models distinguish descending predictions from ascending residuals [1]. Backward masking can preserve early visual activity while disrupting later reentrant components and objective detection [2]. Primate and human studies distinguish feedforward- and feedback-dominant frequency channels in visual hierarchies [3–5]. Hippocampal gamma ranges can be associated with different input routes [6]. Relative phase can change effective interregional influence [7,8]. Thalamocortical recurrence can amplify cortical connectivity and sustain task-relevant activity [9,10]. Hard object-recognition images can recruit later computations that ordinary feedforward models capture poorly [12]. Omission itself can evoke hierarchical prediction responses [13], and population codes can exhibit robust collective modes [14]. Closed-loop brain–machine-interface studies show why online coupled performance cannot be inferred safely from offline decoding alone [15,16].

These findings do not collapse into one binary distinction. “Feedforward” can refer to anatomy, temporal order, a computational graph, or an estimated direction of influence. “Feedback” can convey prediction, gain control, error correction, attention, persistence, or a changed receiver condition. Loss can mean deletion of a stimulus feature, failure of a spike, masking of a later response, corruption of a decoded variable, or failure of a behavioral report. The scientific task is therefore to define a common measurement space in which these mechanisms can be compared without erasing their differences.

The central proposal of this paper is that the relevant object is a **communication regime**: a state-dependent combination of forward propagation, return influence, tolerance of missing events, sensitivity to order, correction, persistence, and receiver transformation. The network need not choose between two globally exclusive modes. Different routes may occupy different points in the regime space at the same time, and the same route may move through that space as ambiguity, surprise, memory demand, damage, or urgency changes.

2. From an engineering question to a biological operation

2.1 The problem came before the name

The source genealogy begins before the 2017 protocol analogy. A 2011 SAN ancestor described a recurrent causal loop in which sensation changes learned neural statistics and prediction; neural

activity changes nerves, muscles, movement, or sound; and the resulting sensory evidence changes the next neural state [20]. In 2012, related sources proposed that learned cellular morphology and synaptic state constrain what a cell recognizes and emits, that small receiver-side differences can alter a larger relation, and that directed graphs and matrices could help describe neural routes [21–23]. These records establish the ancestry of the problem—receiver transformation inside a recurrent sense–action loop—without assigning later terminology to the earlier sources. This genealogy does not backdate NAPOT, PWD, or the ATP name.

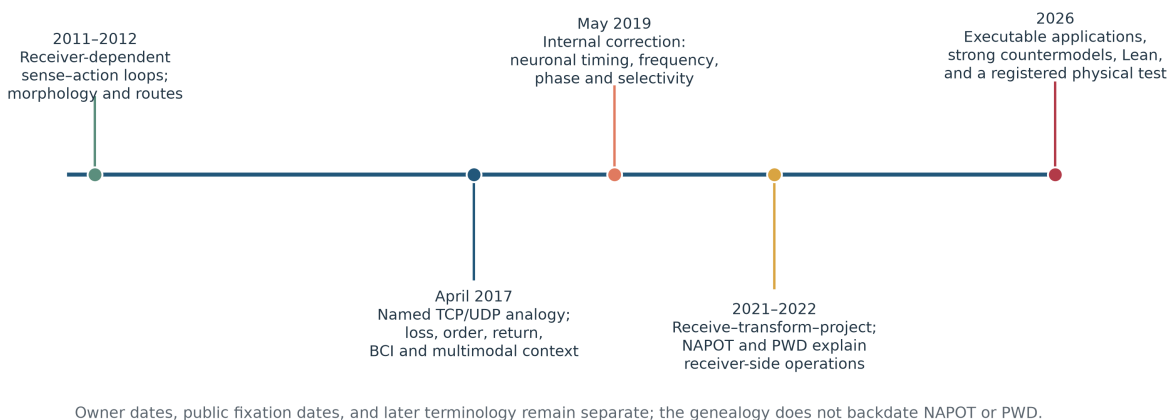
On April 11, 2017, I asked on the *Neural Lace Podcast* whether different kinds of brain communication resemble the functional contrast between Internet transports [24]. The analogy separated traffic that depends on ordering, verification, and a return path from traffic that favors speed and continuity despite missing updates. I proposed that a nervous system might alternate or mix these functional regimes across levels and modalities, and named the proposal **Alternating Transmission Protocol**. Follow-up episodes connected the question to brain–computer interfaces, observer feedback, WebVR/game-streaming completion, and NerveGear-style read/write systems [25–28].

The complete April 2017 context is broader than the slogan “the brain uses TCP or UDP.” The episode moves from dropped frames and local perceptual completion to a multimodal example in which a steak-related pattern would include visual, auditory, and other coordinated properties, then asks how a read/write interface could present a structured pattern to a brain already occupying another state [24]. The question is consequently about *stateful translation*: what may be omitted without destroying a usable relation, what requires ordered return influence, how a receiver changes the arriving relation, and how an artificial interface can determine whether its intervention was incorporated as intended. The source-recovery record preserves that joined problem while separating it from later NAPOT, PWD, and coadaptive-interface terminology.

The May 2019 podcast supplied a particularly important internal technical challenge [29]. The interlocutor rejected a literal packet-size interpretation and redirected the conversation toward what neurons actually transform: presence and absence, firing frequency, timing, phase, and receiver selectivity. That challenge reveals the strongest form of the original proposal. The scientifically useful question is not whether a neuron sends a TCP or UDP datagram. It is whether neural communication occupies distinguishable, dynamically selected regimes of loss tolerance, forward and return influence, ordering, timing, correction, persistence, and consequence.

2.2 Later SAN work fills in the biological middle

Source-recovered development of the ATP research question



Recordings from 2021 described successive neural populations as receiving partial relations, transforming those relations through learned structure, activation, inhibition, timing, and phase, and projecting a changed relation onward [30,31]. In 2022, NAPOT and PWD gave this receiver operation a more explicit form [32,33]. The arriving activity is not a self-contained object copied unchanged through the brain. It changes a receiver that already has a morphology, membrane condition, inhibitory context, recent history, oscillatory relation, and learned tuning. The receiver's subsequent output expresses the difference produced by that encounter.

In this formulation, a PWD is a candidate receiver-relative departure from an expected or maintained relation. NAPOT is the proposed multi-array reconstruction process through which locally transformed relations contribute to a distributed neural rendering. The 2024 NDCA extension asks which relations remain continuously usable as nonlinear receivers transform them across scales [34]. These later concepts do not replace the 2017 question. They explain how a biological regime might work after the packet metaphor is removed.

The resulting division of labor is:

- **PWD:** what informative receiver-relative difference occurred?
- **NAPOT:** how can transformed relations across neural arrays contribute to a reconstructed state?
- **ATP:** how much rapid forward propagation, return influence, repetition, correction, and persistence does the current route recruit?

This division creates testable separations. A PWD-like residual could occur during either a predominantly forward or recurrent regime. A recurrent loop could operate without successfully reconstructing a stable percept. A NAPOT-like reconstruction could use different regime mixtures across sensory conditions. If the three quantities cannot be measured or perturbed independently, the proposed architecture must be revised.

3. What established results contribute

3.1 Forward flow and return influence

Predictive coding provides one established computational decomposition: higher levels predict lower-level activity while forward pathways carry residual error [1]. The decomposition is relevant because “return traffic” need not retransmit the same content. It can change the receiver’s expectation so that a later forward event has a different consequence.

Human masking and TMS studies provide temporal leverage. Fahrenfort and colleagues found that masking could spare an early extrastriate response while removing later reentrant components and reducing objective detection [2]. Hurme and colleagues reported that early V1 perturbation affected both conscious and unconscious visual effects, whereas later perturbation selectively affected conscious vision in their paradigm [11]. These experiments distinguish early and later causal contributions. They do not require every recurrent event to be conscious, nor do they make recurrence alone a complete account of conscious content.

Kar and colleagues approached the problem through recognition difficulty [12]. Some images produced later, behaviorally sufficient inferior-temporal solutions that feedforward models predicted poorly, whereas models with greater recurrent or effective nonlinear depth better captured the response. This motivates a conditional regime: easy inputs may be solved rapidly, while ambiguous or difficult inputs recruit additional transformation.

3.2 Direction, frequency, and phase

Laminar and multi-area recordings show that direction and spectral content can covary. In monkey visual cortex, gamma-associated influence was stronger in feedforward directions while alpha- or beta-associated influence was stronger in feedback directions in the measured tasks [3,4]. Human MEG produced a related gamma-forward and alpha/beta-feedback organization across visual areas [5]. These results are not a universal frequency dictionary. They establish that communication regime can have measurable spectral and directional structure.

Colgin and colleagues found that different hippocampal gamma-frequency ranges were associated with different effective input routes to CA1 [6]. Communication-through-coherence proposes that relative phase controls whether an arriving volley reaches an excitable receiver [7]. Stefanou and colleagues supplied causal human evidence: the phase relationship of motor-cortical mu rhythms changed the efficacy of interhemispheric inhibition induced by paired TMS [8]. Together these results justify including phase and receiver state in the regime model. They also prevent “signal arrival” from being treated as sufficient: the same physical event may have different consequences at different receiver phases.

3.3 Persistence and thalamocortical reinforcement

Mediodorsal thalamic input can amplify local prefrontal connectivity and sustain rule-specific cortical sequences [9]. Reciprocal excitation between anterior lateral motor cortex and thalamus can maintain preparatory activity during a delay; perturbing either node can diminish activity in the other and impair behavior [10]. These results illustrate a loop that does more than echo a message. Repeated interaction changes the stability and effective gain of a cortical state.

This is the biological operation that an abstract “acknowledgement” metaphor can conceal. A return path may focus, amplify, gate, compare, or maintain a relation. The useful variable is therefore not whether a reply packet exists, but how much causal return influence changes accuracy, persistence, calibration, or action.

3.4 Missing events, robust codes, and energetic cost

A missing event can have several meanings. If a representation is redundant, deleting some events may leave performance nearly unchanged. If an event violates an expectation, its omission may produce a large response [13]. Retinal population recordings show that some collective modes can be robust and discriminable under repeated input [14]. The same deletion fraction can therefore be harmless in one code and highly informative in another.

Neural communication is also metabolically constrained. Information transmission carries measurable energetic cost [17], gray-matter signaling has a structured energy budget [18], and activity-dependent local ATP synthesis is required for sustained presynaptic function [19]. The historical acronym’s resemblance to adenosine triphosphate becomes scientifically useful only at this level: energetic availability and cost may influence how much recurrence, repetition, and sustained activity a circuit can afford. Metabolic ATP is a constraint on regime selection, not the semantic message and not the protocol itself.

3.5 Measurement is not yet causal identification

The paper requires a stricter vocabulary for direction. Multivariate Granger-causal analysis asks whether the past of one measured process improves prediction of another after conditioning on a declared set of measured processes [35]. Phase-slope index estimates a frequency-resolved direction from phase progression and was designed to resist particular mixtures of independent sources [36]. Time-reversed Granger checks can reject some directionality produced by mixtures or measurement noise under declared linear-process assumptions [37]. Dynamic causal modelling instead compares explicit generative models in which hidden states, experimental inputs, and hypothesized couplings produce observations [38]. Experimental-design work further shows that the design needed to distinguish a feedback connection can differ from the design needed to detect modulation of a connection already assumed to exist [39].

These are complementary evidence classes, not interchangeable stamps of causality. An observational direction estimate can identify a reproducible predictive asymmetry. A generative-model comparison can identify which declared architecture best explains the measurements under its assumptions. A timed perturbation, matched sham, pathway control, and rescue can support a causal route claim. The causal-inference contract therefore reserves *causal return influence* for the intervention tier and uses *estimated directed influence* everywhere else.

4. Formal model

4.1 Receiver-dependent neural communication

Let the nervous system at time t be a directed, time-varying graph

$$\mathcal{G}_t = (V, E_t), \quad (1)$$

where each node can denote a neuron, population, cortical layer, nucleus, or measured subsystem at the declared scale. Receiver j has state $x_j(t)$, including its measurable membrane, synaptic, inhibitory, neuromodulatory, oscillatory, and recent-history variables. An arriving event from sender i is $e_{ij}(t - \tau_{ij})$. The receiver update is

$$x_j(t + \Delta t) = \Phi_j \left(x_j(t), \sum_i w_{ij}(t) e_{ij}(t - \tau_{ij}), u_j(t), \eta_j(t) \right), \quad (2)$$

where u_j denotes controlled or contextual inputs and η_j denotes unresolved variation. Equation (2) makes the key biological translation explicit: communication is not the transport of a context-free payload. It is a state transition produced when structured input encounters a structured receiver.

For a defined hierarchy or task route, let D_t^+ denote measured forward directed influence and D_t^- measured return influence. These may be estimated observationally, but causal language is reserved for perturbation-based estimates. The recurrent contribution is summarized by

$$\rho_t = \begin{cases} \frac{D_t^-}{D_t^+ + D_t^-}, & D_t^+ + D_t^- \geq \tau_D, \\ \text{NA}, & D_t^+ + D_t^- < \tau_D, \end{cases} \quad (3)$$

where $D_t^+, D_t^- \geq 0$ are estimates from the same declared estimator family and $\tau_D > 0$ is a preregistered identifiability floor. A nearly signal-free window is marked unavailable rather than converted into an apparently precise ratio. A high ρ_t means that measured return influence is large relative to total bidirectional influence at the declared scale. It does not imply that the whole brain is in one feedback state.

4.2 Loss, order, correction, and persistence

Let $Q(\ell)$ be held-out task quality after a controlled fraction ℓ of prespecified input or communication events is removed. Normalized loss tolerance is

$$L(\ell) = \frac{Q(\ell)}{Q(0) + \varepsilon}. \quad (4)$$

The deletion operator must be declared: random spikes, stimulus frames, features, pathway perturbations, or model messages are not interchangeable. The complete deletion curve, rather than one percentage, is the primary result.

For Equation (4), Q must be a nonnegative, preregistered quality score with $Q(0) > \tau_Q$. If the natural endpoint is an unbounded signed statistic, it is transformed to a bounded proper score or replaced with an explicit loss before normalization; adding ε may not conceal a zero-quality baseline.

Let π be a controlled permutation that preserves event identity and count while changing order. Order sensitivity is

$$O_\pi = \frac{Q(\text{id}) - Q(\pi)}{|Q(\text{id})| + \varepsilon}. \quad (5)$$

An alternative distributional endpoint can use Jensen–Shannon divergence between output distributions under original and permuted order. Equation (5) is signed: $O_\pi < 0$ records a prespecified permutation that improves quality and is not clipped away. A regime can be loss tolerant yet order sensitive, or loss sensitive yet relatively order tolerant; these are separate axes.

After perturbation at t_p , let E_{pre} be the error immediately after the perturbation and $E_{\text{post}}(\Delta)$ the error after a recovery interval Δ . Correction gain is

$$G_c(\Delta) = \frac{E_{\text{pre}} - E_{\text{post}}(\Delta)}{E_{\text{pre}} + \varepsilon}. \quad (6)$$

Equation (6) is evaluated only when $E_{\text{pre}} > \tau_E$; otherwise correction gain is unavailable because there was no resolvable error to correct. Negative values are retained as perturbation worsening rather than forced into a rescue narrative.

Persistence after the driving input ends at t_0 is measured by a normalized state or decoder similarity

$$P(\Delta) = \frac{\text{sim}[z(t_0), z(t_0 + \Delta)] - P_{\text{null}}(\Delta)}{1 - P_{\text{null}}(\Delta) + \varepsilon}, \quad (7)$$

where z is the prespecified task-relevant representation and the null accounts for chance similarity and slow drift.

The similarity function, its admissible range, and the drift null are frozen per endpoint. Persistence is not inferred from a slowly varying nuisance signal: the representation must decode the declared task variable above the matched null, and a content-shuffled state supplies a second control.

4.3 Receiver-state dependence and access consequence

Receiver-state dependence asks whether the same class of arriving event has a different consequence because the receiver was in a different prior state. One observational measure is conditional mutual information:

$$S_R = I(Y_{t+\Delta}; X_R^{\text{pre}} \mid E_t, C_t), \quad (8)$$

where Y is the downstream outcome, X_R^{pre} the receiver state before arrival, E_t the arriving event, and C_t measured context. A causal estimate additionally perturbs receiver state while matching the event and context.

The result must then be connected to consequence. Define an access vector

$$\mathbf{A}_t = (A_t^{\text{report}}, A_t^{\text{memory}}, A_t^{\text{action}}, A_t^{\text{calibration}}), \quad (9)$$

rather than reducing access to one verbal report. Different experiments may legitimately use different components, but the endpoint must be declared before analysis.

4.4 The communication-regime vector

At a declared route, scale, and time window, the ATP regime is

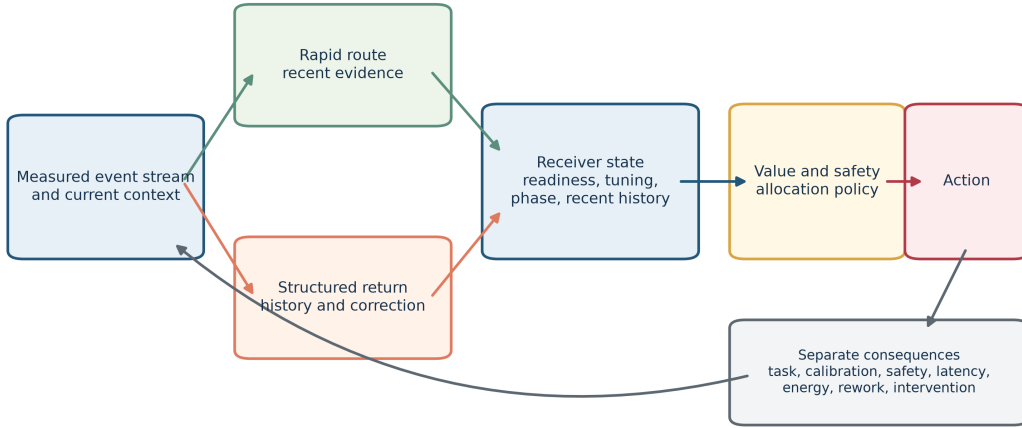
$$\mathbf{r}_t = [\rho_t, L_t, O_t, G_{c,t}, P_t, S_{R,t}, \mathbf{A}_t, E_t, T_t], \quad (10)$$

where E_t is energetic cost and T_t latency. The historical TCP-like and UDP-like descriptions become regions in this space rather than literal biological packet types. A rapid loss-tolerant region is expected to have relatively low return influence, high L , short latency, and limited correction or persistence. A recurrently stabilized region is expected to have higher return influence, correction gain, persistence, and receiver-state dependence, with potential latency and energy costs. Empirical data may reveal additional regions or continuous manifolds.

4.5 A state-dependent gate

Receiver-dependent allocation of rapid and recurrent processing

Candidate regime coordinates: forward influence | return influence | controlled loss | order | correction | persistence | readiness | action coupling | energy | latency



ATP is the proposed allocation policy over these operations; it is not a claim that neurons transmit Internet packets.

Let ambiguity a_t , calibrated uncertainty u_t , estimated event loss ℓ_t , surprise s_t , memory demand m_t , urgency q_t , energetic cost e_t , and latency cost d_t be independently measured or manipulated. A recurrent-allocation gate is

$$p_t = \sigma(\theta_0 + \theta_a a_t + \theta_u u_t + \theta_\ell \ell_t + \theta_s s_t + \theta_m m_t - \theta_q q_t - \theta_e e_t - \theta_d d_t), \quad (11)$$

where σ is a logistic function. The effective computation can be represented initially as

$$\hat{y}_t = (1 - g_t)F(x_t) + g_t R(x_t, h_{t-1}), \quad g_t \sim \text{Bernoulli}(p_t), \quad (12)$$

where F is a forward transformation and R a recurrent transformation with state h . A differentiable relaxation can be used during training, but final evaluation must include discrete-budget conditions so the gate cannot hide unlimited recurrence in a continuous coefficient.

Biological and engineering systems can oscillate pathologically if they switch at every small fluctuation. A hysteretic state rule is therefore

$$g_t = \begin{cases} 1, & p_t + \kappa g_{t-1} \geq h_{\text{on}}, \\ 0, & p_t + \kappa g_{t-1} \leq h_{\text{off}}, \\ g_{t-1}, & \text{otherwise,} \end{cases} \quad h_{\text{off}} < h_{\text{on}}, \quad (13)$$

which provides separate entry and exit thresholds.

The training or model-selection objective is

$$\mathcal{J} = \mathcal{L}_{\text{task}} + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}} + \lambda_E E + \lambda_T T + \lambda_R N_R + \lambda_S N_{\text{switch}}, \quad (14)$$

where N_R is recurrent-compute use and N_{switch} penalizes unstable switching. The ATP hypothesis predicts that an adaptive policy improves a prespecified quality–cost frontier, not merely raw accuracy purchased with more computation.

4.6 PWD, surprisal, and ATP

Let $\delta_{ij}(t)$ be a circular receiver-relative phase departure between an observed receiver relation and the relation expected from sender i , delay τ_{ij} , and receiver context:

$$\delta_{ij}(t) = \text{wrap} \left[\phi_j(t) - \hat{\phi}_{ij}(t \mid x_j(t^-), \phi_i(t - \tau_{ij}), C_t) \right]. \quad (15)$$

Its conditional surprisal is

$$\mathcal{I}_{ij}(t) = -\log_2 p(\delta_{ij}(t) \mid x_j(t^-), C_t). \quad (16)$$

Equation (16) separates statistical rarity from biological importance. A rare departure becomes functionally meaningful only if a tuned receiver detects it and it changes a later state, route, memory, report, or action. A PWD is therefore modeled as a receiver-relative candidate difference with measurable consequence, not as a synonym for generic phase coding.

ATP and PWD answer different questions. PWD supplies a difference variable inside a route. ATP supplies a policy for whether that relation proceeds mainly through rapid forward transformation or recruits repeated and return-path-dependent stabilization. One joint switching value is

$$V_R(t) = \mathbb{E}[\Delta Q_t \mid \mathcal{I}_t, S_{R,t}, a_t, u_t, m_t] - \lambda_E \Delta E_t - \lambda_T \Delta T_t, \quad (17)$$

with recurrence recruited when V_R exceeds the hysteretic entry threshold. This makes the switch responsive to informative difference and receiver state while preserving explicit cost.

4.7 NAPOT and multi-array reconstruction

Across arrays $k = 1, \dots, K$, a minimal transformed-projection sequence is

$$x_{k+1}(t + \Delta) = \Phi_k(x_{k+1}(t), W_{k \rightarrow k+1}x_k(t), B_{k+1}(t), g_{k,t}), \quad (18)$$

where B collects receiver conditions and $g_{k,t}$ controls the local regime mixture. A reconstructed task variable is

$$\hat{z}_t = \Psi(x_1(t), \dots, x_K(t); \mathbf{g}_t). \quad (19)$$

The discriminating question is whether measured or manipulated regime allocation predicts reconstruction quality and perturbation recovery beyond established feedforward, recurrent, predictive-coding, attention, and state-space baselines.

4.8 Identification ladder and hidden-common-cause test

For an observational time series, the first directed-influence estimator is conditional multivariate Granger information in bits per sample:

$$\hat{G}_{i \rightarrow j|C} = \frac{1}{2} \log_2 \frac{\det \hat{\Sigma}_{j|j^-, C^-}}{\det \hat{\Sigma}_{j|j^-, i^-, C^-}}, \quad (20)$$

where the two residual covariance estimates come from the same stable vector-autoregressive fit, i^- and j^- are declared lag histories, and C^- is the preregistered conditioning set [35]. Model order is selected inside each training fold. Stationarity, residual whiteness, stability, and held-out prediction are required; failed assumptions make the estimate unavailable rather than causal.

A latent common source can create apparent direction without a direct edge. For example,

$$X_t = aZ_{t-1} + \epsilon_t^X, \quad Y_t = bZ_{t-2} + \epsilon_t^Y \quad (21)$$

allows past X to predict Y even though intervening on X does not change Y . Equation (21) is the required counterexample behind the hidden-common-cause gate. Every observational ATP analysis therefore adds measured common drivers where available, source-space leakage controls, phase-slope direction [36], time reversal [37], lag-shift and circular-shift surrogates, and leave-one-region-out conditioning. Agreement strengthens an observational direction claim but does not turn it into an intervention.

The causal estimand is instead a controlled contrast,

$$\Delta_{i \rightarrow j}^{\text{do}}(c) = \mathbb{E}[Y_{t+\Delta} \mid \text{do}(X_i = x_1), C = c] - \mathbb{E}[Y_{t+\Delta} \mid \text{do}(X_i = x_0), C = c], \quad (22)$$

with sham intervention, off-route control, matched load, temporal specificity, and rescue where ethically and technically possible. A DCM family can then compare forward-only, return-only, bidirectional, common-driver, and context-modulated architectures [38,39]. The held-out model evidence against baseline b is

$$\Delta \text{ELPD}_{\text{ATP}, b} = \frac{1}{N} \sum_{n=1}^N [\log p(y_n \mid M_{\text{ATP}}, \mathcal{D}_{\text{train}}) - \log p(y_n \mid M_b, \mathcal{D}_{\text{train}})]. \quad (23)$$

ATP reaches the strongest identification tier only when the regime manipulation changes the predicted route-specific outcome, a return-path intervention selectively removes the benefit, and matched rescue restores it without a nonspecific arousal, compute, or stimulation-intensity explanation.

5. Testable predictions

The framework produces eight linked predictions.

1. **Difficulty-dependent recurrence.** Ambiguous, occluded, or otherwise challenging inputs should increase return influence, correction gain, receiver-state dependence, or processing latency relative to matched easy inputs.
1. **Selective loss tolerance.** Some rapid-forward tasks should preserve performance across a controlled deletion range without proportional growth in return influence. Omission-sensitive tasks should show a different curve because the absence itself becomes informative.
1. **Order and loss dissociate.** Event deletion and event permutation should produce separable effects. A system may tolerate missing redundant events while remaining sensitive to temporal ordering.
1. **Receiver-state causality.** Perturbing the pre-event state of a receiver while holding the arriving event approximately constant should change downstream consequence in conditions predicted by Equation (8).
1. **Recurrent rescue.** A targeted disruption should be followed by measurable recovery when the relevant return path is intact; disrupting that path should reduce G_c , persistence, or calibrated performance.
1. **Adaptive-frontier advantage.** Under mixed task demands, a gated model should outperform compute-matched fixed-feedforward, fixed-recurrent, and ungated-mixture baselines on a preregistered quality–latency–energy frontier.
1. **PWD–ATP separability.** Receiver-relative surprisal should predict when recurrence is useful, but comparable PWD values should still occur in more than one regime. If PWD and recurrence are perfectly redundant, the division of labor is unsupported.
1. **NAPOT consequence.** Regime allocation should predict the stability, revision, or recovery of a distributed reconstructed variable. If ATP measures do not improve prediction beyond existing state-space and recurrent models, the NAPOT–ATP bridge is unnecessary.

6. Application proof

6.1 Purpose

The first application is not intended to imitate the whole brain. It tests the paper’s specific engineering claim: a learned gate should allocate recurrence only when recurrence improves the quality–cost tradeoff. The proof must expose its event history, internal state, gate decisions, compute use, outputs, and failures so that every graph can be reproduced from raw data.

6.2 Three matched tasks

Task A: loss-tolerant streaming classification. A sequence contains redundant evidence for a class. During evaluation, prespecified event-deletion schedules remove frames or features. The desired solution is fast and robust without unnecessary recurrence.

Task B: ambiguity-dependent reconstruction. Early inputs are compatible with several classes; a later cue or return computation resolves the ambiguity. Occlusion and adversarially matched distractors prevent a simple confidence threshold from solving the task automatically.

Task C: mixed-demand switching. Trials from Tasks A and B are interleaved with memory-delay and omission trials. The gate must infer the current demand without receiving the ground-truth regime label at test time.

The reference-application contract uses length-24 event sequences with eight output classes and one shared input encoder. Task A draws redundant class evidence across the sequence and evaluates independent and burst deletion at $\ell \in \{0, 0.10, 0.25, 0.40\}$. Task B uses an early ambiguous prefix and a class-resolving cue between positions 17 and 20; a matched distractor carries equal amplitude and marginal frequency without resolving the class. Task C interleaves A and B with four- and eight-step memory delays plus omission trials. The gate receives no ground-truth task or regime label.

All tasks use the same input budget, output classes, encoder width, training examples, and frozen train/validation/test partitions. Data generation seeds 1001–1010 are development seeds; seeds 2001–2020 are untouched final-test seeds. The final split includes two distribution shifts: independent deletion becomes contiguous burst deletion, and the resolving cue moves outside the training-time position range. Test labels remain inaccessible until the configuration, code hash, and analysis plan are frozen.

6.3 Competing models

The benchmark includes:

1. a fixed feedforward model;
2. a fixed recurrent model with matched maximum capacity;
3. a fixed-depth model with compute matched to the recurrent baseline;
4. an ungated mixture using both branches on every trial;
5. a confidence-threshold gate;
6. an ATP gate using Equation (11);
7. ATP without hysteresis;
8. ATP without receiver-state inputs;
9. ATP without energy or latency cost;
10. ATP with shuffled return messages as a causal-content control.

The comparison prevents a recurrent advantage from being confused with parameter count, depth, or unrestricted compute. It also tests whether the gate uses meaningful return content rather than merely adding iterations.

6.4 Required outputs

Each run writes a machine-readable event log containing trial ID, seed, condition, deletion/permutation schedule, ambiguity, uncertainty, branch outputs, gate probability, gate state, recurrent steps, latency, estimated energy/compute, final output, confidence, error, and recovery. The release must include:

- raw run logs;
- a versioned configuration and environment lock;
- tests for event ordering, deletion, gate hysteresis, and metric calculation;
- a system-context diagram;
- a component and data-flow diagram;
- deletion and permutation response curves;
- accuracy–latency and accuracy–energy Pareto plots;
- gate calibration and reliability plots;
- condition-wise recurrence allocation;
- perturbation and matched-rescue plots;
- ablation tables with uncertainty intervals;
- a failure gallery with the exact input and event trace.

The event log is append-only and records `run_id`, code/configuration hashes, data seed, split, task, condition, trial, time index, input event, deletion/permutation flag, encoder state hash, forward output, recurrent input and output, gate probability and discrete state, switch reason, recurrent-step count, latency, multiply-accumulate estimate, confidence, prediction, target, and error. Summary tables are derived only from this log. A graph without a reproducible row-level route back to the log fails the application gate.

6.5 Primary endpoint

Let normalized quality, energy, and latency on held-out trials be $\tilde{Q}$, $\tilde{E}$, and $\tilde{T}$. A preregistered utility is

$$U = \tilde{Q} - \alpha\tilde{E} - \beta\tilde{T}, \tag{24}$$

with α and β frozen before the final run. ATP passes the application gate only if it improves held-out U over every fixed and ungated baseline, the result survives compute matching and seed uncertainty, the gate responds in the predicted direction to task manipulations, and the advantage disappears or changes appropriately in the prespecified ablations.

6.6 Frozen system boundary

The application has seven components: event generator, shared encoder, feedforward branch, recurrent branch with bounded state, regime gate, output head, and append-only evaluator/logger. Only the regime gate may select whether the recurrent branch contributes on a trial. The recurrent branch has a hard maximum of four update steps; the fixed-recurrent and ungated-mixture baselines receive the same branch and maximum compute. Parameter-count matching is reported separately

from realized-compute matching. The implementation contract, estimator ledger, and nonduplication crosswalk were frozen as design commitments before execution.

6.7 Development-seed implementation

The reference application turns the ordinary problem into three small, inspectable sequence tasks. Task A repeats class evidence across 24 events and independently deletes 0%, 10%, 25%, or 40% of them. Task B presents an ambiguous class pair, then supplies one resolving event at position 17, 18, 19, or 20; half the trials add an equal-amplitude cue-like distractor that is inconsistent with the earlier pair. Task C presents one class-bearing event followed by a four- or eight-event delay; the omission control removes that event entirely. Every condition contains all eight classes.

The shared forward branch uses the four most recent events. This gives it a purposeful advantage when redundant current evidence remains available and a purposeful limitation when a decisive event must be integrated over a longer interval. The bounded recurrent branch combines broad accumulated evidence with the strongest class-specific event. It is an interpretable engineering proxy for stateful integration, not a model of a named biological circuit. The gate receives no task identifier, target, regime label, or final correctness signal. It sees only quantities computed from the evolving input: forward uncertainty, top-two ambiguity, observed deletion, event departure, an early-versus-recent state mismatch, inferred memory demand, urgency, and a declared cost term.

All ten systems call the same generator, encoder, forward branch, recurrent branch, and output function. They expose 96 equal declared scalar state/configuration slots and the same maximum of five compute units. The slot count is an implementation-capacity accounting device, not a claim that the analytic reference functions contain 96 learned biological or machine-learning parameters. Realized compute is recorded separately. Fixed feedforward uses one unit; a four-update recurrent decision uses five. The fixed-depth control spends five units but returns the forward output. The shuffled-return control preserves the gate and cost while circularly permuting recurrent class content.

Seeds 1001–1006 calibrated only the two hysteresis thresholds and the confidence comparator’s threshold. The selected ATP thresholds were ($h_{\mathrm{on}}=0.50$) and ($h_{\mathrm{off}}=0.20$); the selected confidence threshold was 0.75. Seeds 1007–1010 were not used for those selections. They supplied 512 base trials and 5,120 system-trials for the development-evaluation table. The final-test seeds 2001–2020 were never generated. Burst deletion and cue positions outside 17–20 remain unopened distribution shifts.

The run produced 307,200 immutable event rows. Each row binds a configuration hash and combined code hash to the seed, split, condition, trial, time index, input event, deletion or distractor status, encoded-state hash, branch outputs, gate probability and state, transition reason, compute, confidence, prediction, target, and error. All tables and figures were regenerated from 12,800 trial-result rows, which include both calibration and development-evaluation trials. Independent validation passed 17 of 17 checks, including exact row counts, 24 events per model-trial, absence of final seeds, identical maximum capacity, exact utility recomputation, manifest hashes, and summary reproduction.

6.8 Development results

Table 1 reports the untouched development-evaluation seeds. Accuracy is included for interpretation, but the frozen primary endpoint is target probability minus 0.08 times normalized energy minus

0.04 times normalized latency.

System	Accuracy	Target probability	Utility	Mean compute
Fixed feedforward	0.6055	0.4134	0.4134	1.0000
Fixed recurrent	0.8906	0.7812	0.6612	5.0000
Fixed depth, compute matched	0.6055	0.4134	0.2934	5.0000
Ungated mixture	0.8867	0.6284	0.5084	5.0000
Confidence-threshold gate	0.8906	0.7663	0.6632	4.4375
ATP gate	0.8906	0.7679	0.6604	4.5859
ATP without hysteresis	0.8906	0.7411	0.6513	3.9922
ATP without receiver state	0.8770	0.7050	0.6241	3.6953
ATP without energy/latency cost	0.8906	0.7743	0.6622	4.7344
ATP with shuffled return content	0.1270	0.1207	0.0131	4.5859

The development result is mixed rather than a pass. ATP exceeded fixed feedforward by 0.24698 utility (95% t interval 0.24531 to 0.24865), fixed-depth compute matching by 0.36698 (0.36531 to 0.36865), the ungated mixture by 0.15200 (0.14891 to 0.15509), ATP without hysteresis by 0.00900 (0.00441 to 0.01359), ATP without receiver state by 0.03626 (0.02739 to 0.04512), and shuffled return content by 0.64721 (0.62938 to 0.66503). These comparisons show that longer integration can matter, compute alone does not reproduce it, content-free return fails, and the implemented receiver-state and hysteresis terms make measurable contributions in this task family.

The decisive adverse result is equally clear. ATP did not exceed fixed recurrence: the paired difference was -0.00086, and its interval (-0.00418 to 0.00246) includes zero. The simpler confidence-threshold gate exceeded ATP by 0.00282; the corresponding ATP-minus-confidence interval was -0.00381 to -0.00184. Removing the energy/latency term changed utility by only -0.00188 with an interval spanning zero. Therefore the full ATP gate has not yet earned an advantage over the strongest simple comparator, and the implemented cost term has not earned an identifiable contribution.

Allocation behavior explains part of the failure. ATP assigned recurrence to every final decision in Tasks B and C, which is directionally appropriate when late disambiguation or memory is required. It also assigned recurrence to 58.59% of Task A final decisions. Accuracy on Task A was already 100%, so this is unnecessary allocation under the present endpoint. The system learned a useful distinction, but not a sufficiently selective one.

The omission control is a second important boundary. With the class-bearing Task C event present, ATP accuracy was 100% at both delays. When that event was removed, accuracy was 6.25% at delay four and 18.75% at delay eight, averaging exactly the 12.5% chance level of an eight-class task. Recurrence preserved available information; it did not recreate target information that the generator never supplied. This negative result is not a defect to hide. It distinguishes persistence from invention and supplies a concrete falsifier for claims that feedback alone guarantees reconstruction.

6.9 Development Study 1 disposition

The executable gate satisfies the integrity and traceability contract but fails the frozen performance pass rule because it does not beat every fixed and ungated baseline. The proper next step is not to open final seeds or retune on seeds 1007–1010. A Development Study 2 revision must first explain, on calibration data or a newly registered development tranche, why confidence alone is sufficient here, define tasks in which receiver state or return content makes a prediction that confidence cannot mimic, and freeze a sharper compute-sparing rule. Only then can a new configuration be evaluated against another sealed development split. The original final-test seeds remain reserved for a genuinely frozen candidate.

6.10 Registered Development Study 2 redesign: confidence is not recurrence value

Development Study 1 earned a narrower design question. A gate that asks only whether the forward branch is uncertain will spend recurrence when no missing relation can be recovered and will miss a confident error that structured return traffic could revise. The redesigned decision variable is the expected utility of recurrence relative to continuing forward:

$$\Delta U_R = U_4(R) - U_4(F), \quad (25)$$

where R and F are the shared recurrent and forward branch outcomes. Equation (25) is available after an outcome for calibration and auditing; it is prohibited as an evaluation-time gate input. The candidate gate instead uses label-blind proxies: historical concentration, historical-versus-recent disagreement, receiver mismatch, evidence absence, urgency, and forward confidence. The target, correctness, task label, declared regime, and realized evaluation benefit remain hidden.

The registered development utility makes rapid-stream recurrence more expensive:

$$U_4 = Q - (0.08 + 0.08q)E - (0.04 + 0.04q)T, \quad (26)$$

where Q is target probability, E and T are normalized recurrent energy and latency, and $q = 1$ for the rapid stream and 0.25 otherwise. This is a software cost model, not a conversion between operations and biological metabolism.

The redesign used four task families and five conditions:

- **rapid redundant:** current evidence is sufficient and urgency is high;
- **irrecoverable low confidence:** the input is uncertain because it contains no class-bearing evidence;
- **confidently wrong:** the latest four events favor a decoy, while earlier structured evidence identifies the target;
- **equal-confidence recoverable:** final forward confidence is exactly 0.125, with earlier target evidence available to return processing;
- **equal-confidence irrecoverable:** final forward confidence is also exactly 0.125, with no earlier class evidence.

Each of eight evaluation seeds supplied 64 base trials, for 512 evaluation trials. Six separate seeds

supplied 384 calibration trials. Twelve systems shared the generator, encoder, forward and return readouts, output rule, 128-slot implementation envelope, and maximum five compute units. The strongest admissible alternative was a ridge-linear benefit predictor trained only on calibration trials from the same label-blind features. A deterministic random gate was matched to the ATP candidate’s 0.375 calibration recurrence rate. A target-informed oracle was reported only as an unreachable analysis ceiling.

The complete run contains 896 base trials, 10,752 model-trials, and 258,048 event rows. Table 2 retains every registered system.

System	Accuracy	Target probability	Utility	Mean compute	Final recurrence
Fixed feedforward	0.3125	0.2639	0.2639	1.0000	0.0000
Fixed recurrence	0.6719	0.6453	0.4728	5.0000	1.0000
Fixed depth, compute matched	0.3125	0.2639	0.0914	5.0000	0.0000
Ungated mixture	0.4219	0.4903	0.3178	5.0000	1.0000
Confidence-threshold gate	0.4219	0.3731	0.2981	3.0000	0.5000
Calibration-trained linear benefit gate	0.6719	0.6198	0.5636	2.5000	0.3750
Rate-matched random gate	0.4551	0.4215	0.3509	2.6172	0.4043
ATP Development Study 2 value gate	0.6719	0.6198	0.5636	2.5000	0.3750
ATP without return-value features	0.6719	0.6198	0.5073	4.0000	0.7500
ATP without cost pressure	0.6719	0.6453	0.4728	5.0000	1.0000
ATP with shuffled return content	0.2969	0.2472	0.1909	2.5000	0.3750
Oracle benefit ceiling	0.6719	0.6198	0.5636	2.5000	0.3750

The revised gate corrected each explicit Development Study 1 allocation failure in the registered fixture. It chose forward processing on every rapid and irrecoverable trial, return processing on every confidently wrong and equal-confidence-recoverable trial, and forward processing on every equal-confidence-irrecoverable trial. The 64 matched pairs had zero forward-confidence difference and a mean recurrence-value gap of 0.87401. Structured return achieved 100% accuracy across the two recoverable conditions, while shuffled return achieved 0%; return content, not extra iterations alone, carried the recovery.

The paired utility intervals favored ATP over fixed feedforward by 0.29969 (0.29968 to 0.29971), fixed recurrence by 0.09082 (0.09076 to 0.09088), fixed-depth matching by 0.47219 (0.47218 to 0.47221), ungated mixture by 0.24579 (0.24555 to 0.24604), the confidence gate by 0.26544 (0.26543

to 0.26546), and the rate-matched random gate by 0.21272 (0.18470 to 0.24075). Removing return-value features reduced utility by 0.05625 (0.05625 to 0.05625). These are favorable development results.

Two null results constrain interpretation. The hand-specified ATP gate and the calibration-trained linear benefit predictor produced identical final decisions and utility. ATP also matched the oracle ceiling exactly in this deliberately separated fixture. The result therefore does not identify the hand-specified ATP score as a unique or necessary implementation; it shows that the relevant observable distinction was learnable and that confidence alone was insufficient here.

One registered adverse result keeps the full Development Study 2 endpoint closed. Removing cost pressure increased recurrence from zero to one across rapid and irrecoverable conditions, but it also raised their mean target probability by 0.04069 rather than remaining within the registered 0.01 ceiling. On rapid trials, accuracy was already 1.0, but target probability rose from 0.80022 to 0.90195. Utility still fell because the declared compute penalty exceeded that sharpening benefit. The failed check reveals that “unnecessary” depends on the quality endpoint: accuracy, target probability, calibration, latency, and energy can disagree. It would be invalid to erase this conflict by changing the endpoint after inspection.

The candidate’s overall 0.671875 accuracy is also not a hidden weakness or a universal benchmark score. The fixture deliberately assigns half of the low-information conditions no target-bearing evidence; balanced eight-class irrecoverable trials remain at chance. All 168 candidate errors occurred inside those retained information-absent conditions. Recurrence was correctly withheld there, but no policy could recover a target that the generator did not encode.

6.11 Development Study 2 disposition

The redesign passes its implementation, separation, allocation, content, baseline, and seal checks but fails the complete registered development endpoint at 20 of 21 checks. It is therefore a **purposeful failed Development Study 2 gate**, not a final application proof. It directly resolves the Development Study 1 confidence-threshold loss and recurrence-overuse defect inside a newly registered fixture, while leaving two stronger boundaries: a learned linear benefit gate is equally sufficient, and probability sharpening prevents a clean claim that all rapid-stream recurrence was quality-neutral.

The next honest gate is not final-seed unsealing. A new preregistered development study must distinguish decision correctness, proper-score calibration, and action consequence; use partially recoverable rather than analytically separable conditions; replace deterministic branch fixtures with learned capacity-matched systems; and test whether ATP-specific variables add value beyond a learned expected-value-of-computation policy. Development Study 1 seeds 1007–1010, Development Study 2 seeds 3007–3014, final seeds 2001–2020, and both original distribution shifts remain unavailable for tuning.

6.12 Registered Development Study 3: learned routes and partial recoverability

Development Study 2’s exact tie with a calibration-trained benefit gate made the next requirement unavoidable. A stronger study could not continue comparing a hand-designed return readout with weak fixed alternatives. Both candidate routes had to learn from the same examples with the same parameter count, and ATP had to exceed the learned expected-value policy rather than merely a

confidence rule.

The registered branch models used two different views of the same 24-event stream:

$$F_\theta(x) = W_F \phi_F(x), \quad R_\theta(x) = W_R \phi_R(x), \quad |W_F| = |W_R| = 264, \quad (27)$$

where ϕ_F contains four two-event bins over the final eight events and ϕ_R contains four six-event bins over the whole sequence. Both representations contain 32 inputs. Both eight-output ridge learners use one intercept per output, the same ridge value, the same 256 branch-training trials, and no evaluation fitting. Capacity equality does not erase the route difference: the forward branch receives high-resolution current evidence, while the return branch receives lower-resolution history.

The generator then supplied four graded conditions: rapid current evidence, temporal conflict, degraded recent evidence, and mixed low confidence. Signal strength, decoy strength, masking, and noise varied continuously. The registered mixture test required each condition to contain at least 15% trials with positive net recurrence value and at least 15% with nonpositive value. This prevents a condition label from serving as an analytically perfect gate.

Classification errors also received different action consequences. With eight classes arranged on a circular action space,

$$d(a, y) = \frac{\min(|a - y|, 8 - |a - y|)}{4}, \quad (28)$$

so a neighboring error receives loss 0.25 and the opposite action receives loss 1. The five separate endpoints were

$$A = \mathbf{1}[\hat{y} = y], \quad C = 1 - \frac{\text{Brier}}{2}, \quad S = 1 - d(\hat{y}, y), \quad G = \mathbf{1}[\text{return}], \quad E \in \{0.2, 1\}, \quad (29)$$

for accuracy, calibration score, action score, recurrence, and normalized cost. The registered policy-selection utility was

$$U_5 = 0.30A + 0.25C + 0.45S - (0.05 + 0.05q)E - (0.03 + 0.03q)T, \quad (30)$$

where urgency q and normalized latency T were prespecified. This composite selected a policy but could not hide a failed component because accuracy, calibration, action, recurrence, and cost each retained an independent gate.

The strongest learned comparator and ATP used the same sixteen standardized, label-blind features and the same ridge coefficients predicting recurrent-minus-forward U_5 . The comparator chose the return branch when the final predicted value was positive. ATP applied that predictor at six checkpoints and added a calibrated hysteretic transition:

$$g_k = \begin{cases} 1, & g_{k-1} = 0 \wedge \sigma(8\widehat{\Delta U}_k) \geq h_{\text{on}}, \\ 0, & g_{k-1} = 1 \wedge \sigma(8\widehat{\Delta U}_k) \leq h_{\text{off}}, \\ g_{k-1}, & \text{otherwise.} \end{cases} \quad (31)$$

Thus ATP received no extra target, feature, coefficient, or evaluation label. Its only possible advantage over learned expected value came from persistent temporal switching.

The complete run contained 1,152 base trials: 256 branch-training, 256 gate-calibration, and 640 untouched development-evaluation trials. Twelve systems produced 13,824 model-trials and 82,944 checkpoint rows. Table 3 retains all evaluation results.

System	Accuracy	Calibration	Action	Recurrence	Cost	U_5
Fixed feedforward	0.5719	0.6025	0.8133	0.0000	0.2000	0.6642
Fixed recurrence	0.8313	0.6340	0.9309	1.0000	1.0000	0.7068
Fixed depth, compute matched	0.5719	0.6025	0.8133	0.0000	1.0000	0.5682
Ungated mixture	0.8094	0.6792	0.9223	1.0000	1.0000	0.7076
Confidence threshold	0.7922	0.6318	0.9148	0.7469	0.7975	0.7168
Learned expected value	0.8531	0.6334	0.9387	0.5609	0.6487	0.7625
Rate-matched random	0.7063	0.6189	0.8754	0.5172	0.6138	0.6869
ATP Development Study 3	0.8531	0.6332	0.9379	0.5344	0.6275	0.7646
ATP without receiver features	0.8219	0.6344	0.9277	0.5453	0.6363	0.7509
ATP without cost pressure	0.8453	0.6355	0.9344	0.7688	0.8150	0.7377
ATP with shuffled return	0.3750	0.5782	0.6547	0.5344	0.6275	0.4799
Oracle net-value ceiling	0.9156	0.6258	0.9684	0.3609	0.4888	0.8114

ATP exceeded fixed feedforward, fixed recurrence, fixed-depth compute matching, ungated mixture, confidence threshold, and rate-matched random gating on registered utility; every ten-seed lower interval exceeded 0.005. The ATP-minus-fixed-recurrence difference was 0.05781 with interval 0.04403 to 0.07159. The ATP-minus-confidence difference was 0.04776 with interval 0.03302 to 0.06250. Structured return mattered sharply: preserving the gate while shuffling return content reduced action score by 0.28320 and utility by 0.28464.

Three of the four conditions realized the registered partial-recoverability mixture. Positive recurrence value occurred on 57.50% of degraded-recent, 61.25% of temporal-conflict, and 24.38% of mixed-low-confidence trials. The rapid condition did not pass: only 1.25% had positive recurrence value, while 98.75% were nonpositive. The learned branches themselves were capacity matched and useful. Forward accuracy reached 98.75% on rapid evidence; return accuracy reached 97.50% on degraded recent evidence.

6.13 Development Study 3 disposition

Development Study 3 passes 24 of 28 registered checks and fails the complete gate. ATP’s mean U_5 exceeded the learned expected-value gate by 0.00206, but the paired interval was -0.00073 to 0.00485 rather than clearing the registered 0.005 lower margin. Accuracy was identical. Calibration difference was -0.00014. Action-score difference was -0.00078, with interval -0.00422 to 0.00265, so

ATP did not earn its registered action superiority. ATP did save 0.02656 recurrence and 0.02125 cost, satisfying the material-cost alternative, but savings alone could not substitute for the failed effectiveness gates.

The receiver-feature ablation also remained unresolved. ATP’s mean utility exceeded it by 0.01364, and exact accuracy rose by 0.03125, but the ten-seed utility interval ranged from -0.00128 to 0.02856 rather than clearing the 0.0025 lower margin. The study therefore cannot attribute the candidate’s behavior specifically to the receiver-relative feature subset.

Removing cost pressure increased recurrence by 0.23438 and cost by 0.18750, while lowering utility by 0.02687. It changed accuracy by -0.00781, calibration by +0.00222, and action score by -0.00352. This is stronger separation than the Development Study 2 cost null, but it does not erase Development Study 2’s target-probability sharpening result. The correct conclusion is endpoint-specific: more recurrence can alter confidence or calibration without improving the action that matters, and cost must be evaluated against the declared use case.

7. Biological experiment program

7.1 Minimum viable human experiment

A first noninvasive study can combine a visual task, controlled occlusion or masking, working-memory demand, EEG/MEG directed-influence analysis, and time-specific TMS. Conditions vary ambiguity, omission, and urgency independently. The preregistered analysis estimates early forward influence, later return influence, spectral directionality, phase relation, performance, confidence, reaction time, and perturbation recovery.

The critical comparison is not “conscious versus unconscious” in the abstract. It is whether matched trials occupy different positions in the regime vector and whether a targeted perturbation changes the specific component predicted to matter. For example, late return-path perturbation should affect ambiguity resolution or persistence more strongly than easy rapid classification, while an early perturbation may affect both.

7.2 Circuit-level experiment

A circuit experiment can record from a defined sensory–association–thalamic route during matched easy, ambiguous, omission, and delay trials. Temporally precise perturbations separately target the forward path, return path, and receiver-state preparation. Matched rescue reintroduces patterned influence with controlled phase and delay. The experiment measures Equation (3) through Equation (9) at one declared scale.

This design distinguishes several mechanisms that verbal “feedback” can blur:

- a return path carrying prediction;
- a return path changing receiver gain;
- recurrence maintaining activity after input offset;
- recurrence correcting a perturbed estimate;
- phase alignment changing effective influence;
- omission generating a new error signal;

- a general arousal change that improves all conditions nonspecifically.

7.3 Closed-loop interface test

For a BCI, stimulation policy should be selected online from measured task state and receiver response. A fixed open-loop stimulation schedule is compared with a return-sensitive adaptive policy. Exact reinstatement is tested against mismatched timing, frequency, phase, location, and intensity controls. The relevant endpoint is not whether a participant reports “something.” It is whether the intended perceptual, motor, or memory variable changes selectively, reproducibly, and reversibly while safety limits remain satisfied.

8. Falsification and model revision

ATP would lose explanatory value under any of the following outcomes:

1. A fixed regime predicts held-out neural and behavioral data as well as the full regime vector after complexity correction.
2. Return influence does not vary with ambiguity, uncertainty, persistence demand, perturbation, or task context in the proposed circuits.
3. Controlled loss and controlled permutation fail to produce separable response profiles.
4. Receiver-state measures add no predictive or causal information once the arriving event and ordinary covariates are controlled.
5. A gated application gains only by using more parameters, more iterations, or leaked task labels.
6. Shuffled or content-free recurrence performs as well as structured return influence.
7. PWD-like receiver-relative differences cannot be identified independently of generic signal amplitude, phase coding, prediction error, or recurrent activity.
8. ATP variables fail to improve reconstruction, recovery, memory, report, or action prediction beyond established baselines.
9. Energetic effects disappear after activity and compute are matched, eliminating the proposed cost constraint.
10. The gate is unstable, poorly calibrated, or systematically selects expensive recurrence where it provides no benefit.

A failed prediction identifies the level requiring revision: measurement, route definition, gate input, regime geometry, PWD coupling, NAPOT coupling, or the broader ATP hypothesis. The source genealogy remains a record of how the question developed; empirical outcomes determine which present formalization survives.

Development Study 1 activates this rule rather than treating it as ceremonial. Outcomes 1 and 10 are provisionally observed in the synthetic development application: fixed recurrence is equivalent within the four-seed interval, the confidence comparator is better, and ATP over-allocates recurrence on an easy rapid-stream task. Outcome 6 is not observed because shuffled return content fails sharply. Outcome 4 is not observed in this implementation because removing receiver-state inputs reduces utility. These are properties of the declared engineering test, not direct biological findings.

9. Discussion

9.1 The proposal is a policy over mechanisms

ATP does not compete with predictive coding, recurrent processing, thalamocortical amplification, communication through coherence, robust population coding, or closed-loop control as if only one can be true. Those traditions identify mechanisms and measurements that may occupy the regime space. ATP adds a policy question: under what conditions does a route rely primarily on rapid transformation, and under what conditions does it recruit return influence, repeated comparison, persistence, and correction?

This policy framing matters because a universal instruction to maximize recurrence would be as implausible as a universal instruction to eliminate it. Recurrent computation costs time and energy and can amplify errors as well as correct them. Purely forward processing can be fast but brittle under ambiguity, damage, or state mismatch. The useful architecture allocates recurrence conditionally and makes that allocation measurable.

9.2 A continuous space, not two boxes

The historical Internet analogy generated a productive contrast, but the biological model is deliberately multidimensional. A route can be highly recurrent and loss tolerant. It can be forward-dominant yet order sensitive. It can have strong return influence without persistent activity, or persistence without accurate correction. Regime clusters should therefore be discovered and cross-validated rather than imposed from two labels.

The word “alternating” is retained because the central claim concerns changing allocation. The change may be discrete, continuous, rhythmic, hierarchical, or route-specific. Equation (13) gives one testable discrete approximation with hysteresis; future data may favor a manifold, oscillator, or control-law representation.

9.3 Why receiver state is central

The same incoming event can facilitate, inhibit, redirect, or fail to affect a receiver depending on its state. This is the bridge from transmission to computation. It connects the 2012 morphology and receiver-difference precursors to the 2019 timing/phase red team and the later PWD formulation. The scientific model therefore tracks not only what arrives, but what the receiver was prepared to recognize and what changed afterward.

This receiver-relative account also clarifies why a signal can be statistically surprising yet biologically irrelevant. Shannon surprisal describes improbability under a model. Meaningful neural consequence additionally requires tuned reception and a change that propagates into memory, report, action, or future inference. Equation (16) is therefore paired with Equations (8) and (9).

9.4 Implications for neural interfaces

A read/write interface cannot assume that the same stimulation pattern has a fixed meaning independent of receiver state. It must estimate whether the target circuit is receptive, whether forward evidence is sufficient, whether a return path is stabilizing or distorting the state, and whether

a perturbation produced the intended consequence. This motivates an interface that alternates among sensing, tentative stimulation, confirmation, correction, and low-overhead continuation.

The longer-term NerveGear question is therefore not merely how to inject an image, sound, taste, or body state. It is how to establish a co-adaptive route whose transformations remain interpretable as the biological network learns and drifts. ATP supplies the regime-control layer; PWD supplies a receiver-relative event description; NAPOT supplies the reconstruction hypothesis; and NDCA supplies the continuity question across nonlinear change.

9.5 Energetic interpretation

The acronym ATP was historically chosen partly because adenosine triphosphate is the cellular energy currency. The present formulation gives that association a disciplined place. Recurrence, sustained firing, synaptic vesicle cycling, and corrective exchange incur costs. A biological controller should allocate those costs where they improve behavior, persistence, or calibration. This produces an empirical energy–accuracy–latency frontier. A direct biochemical coupling would require local energetic measurements and causal manipulation; the current model requires only that energetic cost be measured as a candidate constraint.

9.6 Nonduplication boundaries

ATP owns the **allocation policy** over rapid forward transformation, return influence, loss tolerance, ordering, correction, persistence, and cost. The PWD calculus paper owns the full receiver-relative event operator and its calculus. The Gamma–Shannon–PWD paper owns conditional self-information relative to tonic expectation. The NAPOT methods paper owns multi-array reconstruction and observability. *The Flow of Information in the Brain* owns the larger laminar, corticothalamic, sensorimotor traffic architecture. *The Coadaptive Substrate Principle* owns bounded query/read/write contracts and long-term coadaptation. Neural Tuning owns learned receiver selectivity. ATP may use each as an input or consequence, but it passes only if its regime-allocation variables and gate add held-out explanatory or engineering value beyond those component accounts.

9.7 What the development failure teaches

The reference application supports a narrower conclusion than the complete ATP hypothesis. Selective access to a longer-lived state can solve tasks that a four-event forward window cannot, and structured return content matters. Yet the complete hand-designed gate does not outperform a confidence threshold. The current task family makes uncertainty highly diagnostic of memory demand, so receiver-state and cost variables have too little independent work to do. A stronger test must decorrelate confidence from the need for recurrence: some low-confidence trials should be irrecoverable and therefore not worth extra compute; some high-confidence trials should be confidently wrong until a receiver-state-dependent return path revises them; and equal-confidence trials should differ in whether structured return content helps. That is a design requirement produced by the result, not an after-the-fact favorable reinterpretation.

9.8 Confidence, value, and the strongest countermodel

The paired equal-confidence design shows why uncertainty cannot serve as a universal recurrence rule. Two streams can have the same current confidence while differing completely in whether an earlier structured relation exists. Conversely, a low-confidence state can be irrecoverable because no class-bearing evidence entered the system. ATP’s source-faithful problem is therefore better expressed as conditional value of return influence than as uncertainty maximization.

The strongest countermodel matters just as much. A calibration-trained linear benefit predictor reproduced every final ATP decision in Development Study 2. That equality is not evidence against dynamic allocation; it is evidence that the present fixture does not distinguish the hand-specified ATP score from a simpler learned expected-benefit policy. Any future claim of an ATP-specific contribution must exceed this learned alternative under distribution shift, partial observability, and capacity matching.

The failed cost-pressure check also refines the measurement problem. Recurrent computation can leave categorical accuracy unchanged while sharpening target probability. If the application values calibrated probability, that sharpening may be useful; if it values a rapid correct action, it may be waste. Energy, latency, calibration, and consequence must therefore be measured separately before being combined. A composite utility can make a decision rule explicit, but it cannot make the underlying tradeoff disappear.

9.9 What learned branches and the expected-value countermodel establish

Development Study 3 replaces a weak version of the comparison with its strongest operational form. The forward and return routes have equal learned capacity. The expected-value comparator sees the same features and uses the same predicted benefit as ATP. The candidate cannot win by receiving a hidden task identity or a handcrafted oracle feature; it can win only if temporal entry and exit state adds held-out value.

The result is informative because ATP did not earn that effectiveness claim. Hysteresis reduced recurrence and cost relative to the learned expected-value gate, but exact accuracy remained equal and action score was slightly lower. The utility interval included zero and stayed below the prespecified superiority margin. A direct learned value decision is therefore sufficient for the present distribution. This does not erase the source-level ATP question. It identifies a strong causal countermodel for the current implementation: conditional expected value without persistent switching.

At the same time, the broader allocation question survives. ATP beat fixed forward processing, fixed recurrence, a confidence threshold, ungated mixing, fixed-depth compute, and rate-matched randomness. Shuffling return content destroyed much of the action and utility gain. These results show that learned content and conditional allocation matter in the registered fixture. They do not show that the ATP-specific hysteretic operator is uniquely required.

The failed rapid-condition mixture is equally important. Three conditions contained overlapping positive and nonpositive recurrence values, but recent target evidence made the rapid condition almost uniformly unfavorable to return processing. A later study should not alter that observed generator after the fact. It should register a new naturalistic or independently sampled workload in which urgency and recurrence value vary within every surface category, then test generalization without using a condition label.

10. Limitations

Development Study 3 adds learned branches but remains a deterministic synthetic development benchmark. Equal coefficient counts do not guarantee equal inductive bias: recent two-event bins and whole-sequence six-event bins expose different information and temporal resolution. Ridge learners are interpretable capacity-matched baselines, not the strongest possible neural architectures. The gate-calibration set fits both benefit coefficients and thresholds, and ten evaluation seeds remain a limited independent sample. Three partial-recoverability mixtures passed, but the rapid condition failed its registered mixture check. The candidate’s receiver-feature advantage was not conclusive, and its expected-value superiority and action-superiority checks failed. Those failures prohibit final-test unsealing and any claim that hysteresis is necessary. The circular action geometry and utility weights are declared engineering choices; other applications require independently justified consequences and costs.

The completed application is a deterministic synthetic development benchmark, not a final benchmark and not a new biological dataset. Its analytic branch functions are designed to expose the regime-allocation question; they are not trained neural networks, measured neurons, or a realistic metabolic model. The four development-evaluation seeds provide only four independent seed-level units for uncertainty intervals. The final-test seeds and distribution shifts remain sealed because the present candidate failed before that gate. “Energy” and “latency” are normalized compute proxies here; software operations, electrophysiology, metabolism, and behavior require separate declared measurements and cannot be converted by analogy.

Directional-connectivity estimates can be sensitive to preprocessing, volume conduction, reference choice, model order, hidden common causes, nonstationarity, and measurement delay; perturbation is required for causal interpretation. The estimator ladder reduces specific failure modes but cannot prove that every latent driver has been measured. Conscious access is multidimensional, and report, memory, action, and confidence cannot be substituted for one another without justification.

The Development Study 2 generator deliberately creates clear recoverable and irrecoverable cases. This makes identifiability visible but makes the allocation boundary easier than natural data. The hand-specified ATP gate and the learned linear benefit gate both reached the oracle ceiling, so no architectural uniqueness is identified. The target-informed oracle is an analysis ceiling, not an admissible deployable comparator. The rapid-stream cost conflict further shows that target probability and action sufficiency are not interchangeable. Eight evaluation seeds improve on Development Study 1’s four-seed interval but remain a small synthetic sample with deterministic branch functions.

The genealogy also contains different grades of date evidence. Public audio and platform records provide strong public anchors for the named 2017 proposal. Some 2011–2012 wording retains owner-asserted dates with later Git fixation. The manuscript uses those sources to describe conceptual ancestry while keeping date class, public fixation, and terminology distinct.

11. Conclusion

Neural communication is not one uniform act of sending. Some relations can move forward rapidly despite incomplete events. Others require return influence, comparison with receiver state, repeated exchange, persistence, or correction. The Alternating Transmission Protocol names the hypothesis that nervous systems dynamically allocate those operations across route, scale, and task.

The first executable study made the proposal answerable and allowed it to fail: the initial ATP gate did not beat fixed recurrence or a confidence threshold and spent recurrence too often on a rapid stream. The registered redesign then separated uncertainty from recurrence value. It withheld recurrence from low-confidence streams with no recoverable information, recruited structured return for confidently wrong streams, and selected opposite policies for trials with exactly equal forward confidence. It beat the original confidence gate and all fixed, ungated, and random baselines on the registered utility while using 2.5 rather than five mean compute units.

Those favorable results do not close the scientific question. A calibration-trained linear benefit predictor exactly matched the ATP candidate, so the fixture does not identify a unique ATP implementation. Removing cost pressure also sharpened probability on already correct rapid trials, violating one registered check and exposing a conflict among accuracy, calibration, latency, and compute. Development Study 2 therefore closes as a 20-of-21 purposeful development failure with a better mechanism, a stronger countermodel, and a more precise next experiment. It is not a neural result, a biological causal proof, or a final benchmark. Final seeds 2001–2020 and both registered distribution shifts remain sealed.

11.1 Development Study 3 conclusion

The next study made both computation routes learned, equalized their parameter counts, graded the available evidence, separated categorical accuracy from calibration and action consequence, and confronted ATP with its strongest learned countermodel. The resulting ATP policy was selective and efficient. It preserved structured return when it mattered, beat all broad fixed and confidence policies, and used less recurrence and cost than direct learned expected value.

Efficiency did not establish the registered ATP-specific claim. The learned expected-value gate matched ATP’s accuracy, slightly exceeded its action score, and fell within the utility interval. One intended partial-recoverability condition was not realized, and receiver-relative feature necessity remained unresolved. Development Study 3 therefore closes as a 24-of-28 purposeful development failure. It strengthens the application and narrows the hypothesis without opening the final benchmark. Final seeds 2001–2020, Development Study 1 and Development Study 2 evaluation seeds, and both registered distribution shifts remain sealed.

The next honest gate is external rather than another adjustment to these observed synthetic seeds: a newly registered workload or measured dataset, a nested calibration design, stronger nonlinear capacity-matched route learners, and action costs derived from the task itself. ATP-specific temporal state should advance only if it adds held-out action or calibration value beyond a learned expected-value-of-computation policy, not merely if it saves operations while preserving an average score.

12. The problem before the protocol name

A navigation system receives a changing stream of partial evidence. Sometimes the latest measurements already identify a safe action. Sometimes those measurements are occluded and an earlier pattern can repair them. Sometimes the world has changed, making old evidence dangerous. Sometimes the evidence arrives in bursts, so a longer view is useful only on a subset of trials. The scientific problem is not whether recurrence exists. It is whether a controller can identify when the additional history will improve the action enough to justify delay and computation.

That problem has four separable parts:

1. learn what the recent evidence supports;
2. learn what the longer history supports;
3. estimate the consequence of choosing one route rather than the other;
4. determine whether persistent temporal switching adds value beyond a direct estimate of expected consequence.

Only after those operations are explicit is the ATP name useful. In this experiment, ATP denotes the fourth operation: a hysteretic state that can enter or leave the return route as predicted value changes over six checkpoints. It is not a synonym for recurrence, confidence gating, phase coding, or feedback in general.

13. A task in which action has a physical meaning

13.1 Safe corridors rather than abstract class distance

The workload contains eight possible steering directions. The correct label is the center of a safe corridor, but an adjacent direction may remain safe when the corridor is broad. Conversely, a small error can be costly when the corridor is narrow and the hazard is severe. If the circular class offset is d , normalized angular error is $a = d/4$, safe half-width is w , and hazard severity is h , movement loss is

$$L_{\text{move}} = (1 - h)a^2 + h\mathbf{1}[d > w]. \quad (32)$$

The first term measures lost progress. The second represents leaving the safe corridor. This makes exact classification and successful action different endpoints rather than two names for the same score.

13.2 Delay belongs to the task

Forward selection requires one registered route tick and return selection requires three. A trial supplies a deadline slack s and urgency u . Delay loss is

$$L_{\text{delay}} = u \frac{\max(0, \ell - s)}{3}, \quad Q = 1 - \min(1, L_{\text{move}} + L_{\text{delay}}), \quad (33)$$

where Q is action score. Recurrence can therefore improve the chosen direction while still reducing action value by arriving too late. The system cannot claim a benefit by treating more computation as free.

13.3 Four measured regimes

Each of 27 new seeds generated 64 trials, sixteen per condition. Clear-current corridor trials were registered as a recurrence-negative control. Recoverable occlusion degraded recent evidence while varying whether earlier evidence was useful or misleading. Late route reversal made a previously

strong route stale. Intermittent navigation alternated evidence bursts and gaps. These names describe generator conditions; they were not available to the gate as input labels.

The split was frozen before outcomes:

- seeds 5001-5006 trained route learners;
- seeds 5007-5009 selected route architecture;
- seeds 5010-5013 trained benefit models;
- seeds 5014-5015 selected benefit architecture;
- seeds 5016-5017 calibrated policies;
- seeds 5018-5027 formed the untouched development evaluation.

14. Nonlinear routes and nested selection

14.1 Equal-capacity nonlinear route learners

Both routes began with 32 standardized measurements. The forward representation used four two-event bins over the latest eight events. The return representation used four six-event bins over all 24 events. Using the random-feature method as a declared nonlinear approximation [40], for standardized input z , deterministic projection W , bias b , and hidden width H , the nonlinear representation was

$$\phi_H(z) = \begin{bmatrix} 1 & z^\top & \tanh(z^\top W + b)^\top \end{bmatrix}^\top. \quad (34)$$

Ridge output weights were learned as

$$\hat{B} = (\Phi^\top \Phi + \lambda R)^{-1} \Phi^\top Y, \quad \hat{p} = \text{softmax}(\phi_H(z)^\top \hat{B}). \quad (35)$$

Hidden widths 32, 64, and 96 and ridge values 0.01, 0.1, and 1.0 were compared on a branch-selection split. One joint pair was selected from the mean forward-return multiclass Brier score, so both routes retained identical declared capacity. The winner used width 96 and ridge 0.01. Each route contained 1,032 fitted output coefficients and the same number of fixed projection parameters. Equal counts do not make the representations equivalent; they prevent one route from winning merely because it received more fitted capacity.

14.2 Nested benefit and policy selection

At six checkpoints, the controller measured eighteen label-blind features covering forward confidence, entropy, signal strength, ordering, historical concentration, history-recent mismatch, evidence persistence, hazard, corridor width, deadline, and delay exposure. A nonlinear benefit model predicted realized return-minus-forward utility. Width and ridge were selected on seeds never used to fit the candidate models; width 32 and ridge 1.0 won. The selected model was then refit on the combined benefit-training and benefit-selection splits.

A trajectory countermodel received eighteen summaries of the complete six-checkpoint benefit path. Policy thresholds were calibrated only after the route and benefit architectures were fixed. This

nested separation was required to reduce model-selection bias [41,42]. The final expected-value gate used the final predicted benefit. The trajectory gate learned from the complete path. ATP used the same checkpoint estimates but applied hysteresis. Thus ATP’s distinctive test concerned persistent switching state, not privileged evidence.

14.3 Separate endpoints and registered utility

The study reported exact accuracy A , calibration score $K = 1 - \text{Brier}/2$, task action score Q , recurrence rate R , measured compute C , and latency separately. Executed matrix operations were counted from the selected architectures and normalized by the registered maximum. The selection utility was

$$U_6 = 0.20A + 0.20K + 0.55Q - 0.05C. \quad (36)$$

This utility made policy selection reproducible. It did not authorize a pass when a registered component failed.

15. Results

15.1 Scale and custody

The complete workload contained 1,728 base trials across all development splits. The untouched evaluation contained 640 base trials, 8,320 model-trials across thirteen systems, and 49,920 append-only checkpoint rows. The accepted run preserved the frozen registrations, code hashes, selected weights, thresholds, trial-level outputs, figures, and every registered check. No prior evaluation seed, final seed, or registered distribution shift appeared in the evidence.

15.2 Candidate performance

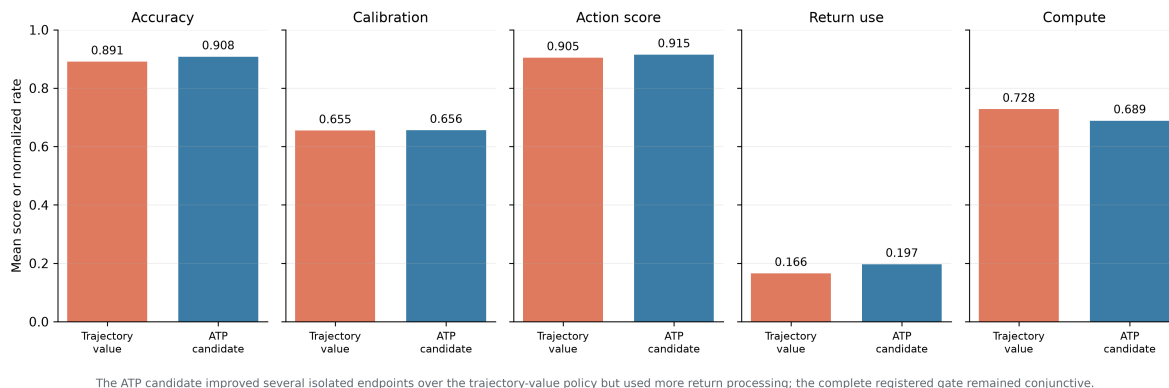
ATP candidate produced:

- exact accuracy: 0.9078125;
- calibration score: 0.6558633;
- action score: 0.9149354;
- recurrence: 0.196875;
- normalized compute: 0.6885747;
- normalized latency: 0.4645833;
- registered utility: 0.7815209.

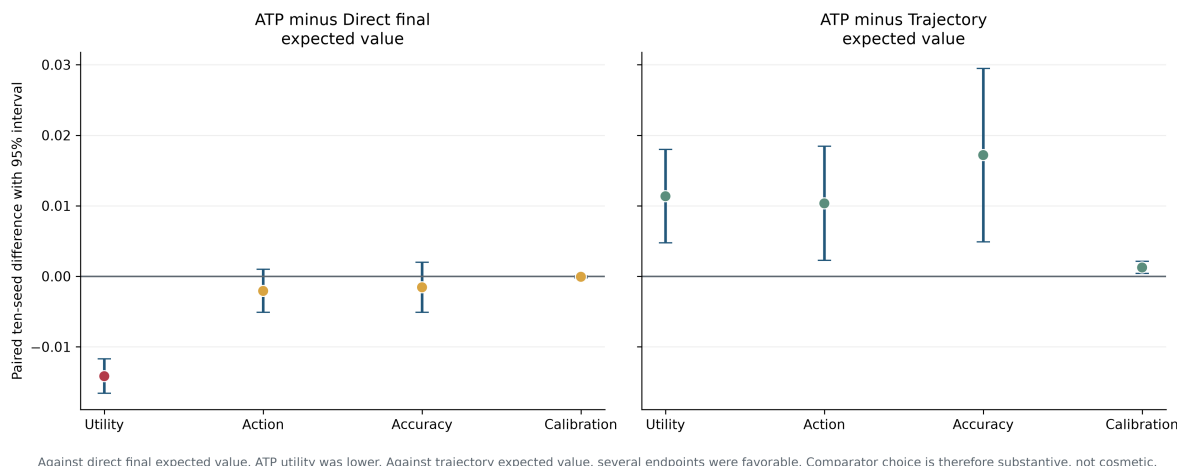
The candidate exceeded the preregistered trajectory expected-value gate in mean utility by 0.011352, with interval [0.004731, 0.017973]. It also improved accuracy by 0.017188 and calibration by 0.001244. Mean action improved by 0.010338, but its interval [0.002238, 0.018439] began just below the registered 0.0025 action margin. ATP selected recurrence 0.03125 more often than the trajectory gate, exceeding the maximum permitted difference, while using 0.03959 less normalized compute because it avoided the trajectory model’s additional evaluation.

15.3 The empirically strongest gate was simpler

Registered nonlinear navigation study: separate endpoints



The strongest countermodel changes the interpretation



Although the trajectory gate was preregistered as the architecturally strongest countermodel, the final expected-value gate performed best among admissible policies. It achieved 0.909375 accuracy, 0.917013 action score, 0.655937 calibration, and 0.795689 utility while using 0.434615 normalized compute. Relative to it, ATP lost 0.014168 utility with interval $[-0.016608, -0.011727]$, lost 0.002078 action, and used 0.253959 more compute. A direct final estimate of expected return value was sufficient for this distribution.

Fixed forward processing also remained competitive at 0.784319 utility. ATP's mean difference was -0.002798 with interval $[-0.019137, 0.013542]$. ATP improved exact accuracy and calibration, but its additional action and recurrence costs did not establish the registered net advantage.

15.4 Structured return and receiver state remained consequential

Shuffling the selected return content reduced action score by 0.052219 and utility by 0.062657. The returned route was therefore not useful merely because it consumed extra operations. Removing

receiver-state features increased recurrence from 0.196875 to 0.476563 and reduced utility from 0.781521 to 0.740879. The paired receiver-feature utility interval cleared its registered margin. Removing task-cost features reduced recurrence to 0.09375 and utility to 0.762088. These ablations support the narrower claim that content, receiver state, and task consequence matter to allocation in this fixture.

15.5 Recoverability was condition specific

Recoverable occlusion realized the required mixture: 25.625% of trials had positive recurrence value and 74.375% had nonpositive value. Clear-current corridor behaved as a clean negative control with zero positive trials. Late reversal was nonpositive on 97.5% of trials. Intermittent navigation did not realize its registered partial mixture: only 1.875% of trials had positive recurrence value. This is a workload failure as well as a model result; it cannot be repaired by renaming the condition after evaluation.

16. The seven failed checks

The conjunctive endpoint passed 24 of 31 checks. Seven failed:

1. **Action superiority over trajectory expected value:** interval lower bound 0.002238, below 0.0025.
2. **Recurrence excess over trajectory expected value:** 0.03125, above 0.01.
3. **Utility superiority over fixed forward:** interval lower bound -0.019137, below 0.005.
4. **Utility superiority over fixed-depth compute matched:** interval lower bound -0.004205, below 0.005.
5. **Utility superiority over rate-matched randomness:** interval lower bound 0.001318, below 0.005.
6. **Utility superiority over final expected value:** interval lower bound -0.016608, below 0.0025.
7. **Intermittent partial recoverability:** positive fraction 0.01875, below 0.20.

No favorable subset converts 24 of 31 into a pass. The full registered endpoint is `FAIL_COMPLETE_DEVELOPMENT_GATE`.

17. Source-faithful interpretation

17.1 What failed

The present hysteretic implementation failed to establish that persistent temporal state adds value beyond direct learned expected value on this navigation distribution. That is the strongest warranted conclusion. The final gate did not need memory of its own entry and exit state; it could act on the last learned estimate and perform better with less computation.

17.2 What did not fail

The historical ATP proposal is broader than this one controller. It asks how a nervous system allocates rapid transformation, return influence, order sensitivity, correction, persistence, receiver preparation, action consequence, and resource constraint. The experiment confirmed that structured return sometimes helps, that the value varies across trials, and that receiver-state and task-cost features change allocation. It did not identify hysteresis as the necessary allocation operator.

This distinction is the anti-strawman boundary. Treating the failed controller as the full historical claim would compress a multidimensional research question into one implementation. Treating the favorable ablations as proof of biological ATP would make the opposite error. The Apex interpretation holds both layers at once: the source proposition is preserved in its strongest context, while the current executable mechanism remains answerable to its preregistered evidence.

17.3 The Development Study 1-Development Study 3 record remains active

Development Study 2’s exact tie with a learned gate remains evidence that a simpler conditional-value policy can reproduce ATP decisions. Development Study 2’s failed cost-neutrality check remains evidence that extra recurrence can sharpen a probability without changing categorical action. Development Study 3’s four failures remain evidence that learned branches and lower compute do not by themselves establish ATP-specific action value. Development Study 4 adds a fourth constraint: nonlinear routes and task-derived consequences still do not guarantee that hysteretic persistence is useful.

These are cumulative constraints, not discarded prototypes. Any later mechanism must explain all of them without retuning the observed seeds.

18. Limitations

The navigation workload is deterministic and synthetic. Its safe-corridor loss is more task grounded than a generic class distance, but it is not a robot, animal, patient, or measured neural circuit. Random-feature ridge models are nonlinear and capacity matched by declared parameter counts, yet they do not exhaust modern sequence learners. Operation counts approximate software computation and do not measure biological energy. The split design protects evaluation custody, but ten evaluation seeds remain a limited independent sample.

The trajectory countermodel was designated strongest before outcomes because it used the full checkpoint path; the simpler final gate then performed better. This is a useful empirical result, but it also shows that architectural complexity is not synonymous with countermodel strength. Intermittent navigation did not realize the intended recurrence-value mixture. That failed workload check reduces the scope of conclusions about intermittent evidence and must remain visible.

The experiment does not measure oscillatory phase, PWD, thalamocortical traffic, conscious access, or biological action selection. It supplies an engineering discriminator for a regime-allocation policy. Translation requires independently justified neural observables, perturbation, receiver-state measurement, and task-specific energetic and behavioral costs.

19. Development Study 4 conclusion

Development Study 4 satisfies the requested methodological upgrade: a new workload, new seeds, nested model selection, nonlinear capacity-matched branches, task-derived action costs, a learned trajectory countermodel, separate endpoints, adverse-result retention, deterministic replay, and intact final seals. Those improvements make the negative result more valuable because the gate can no longer be blamed on the linear routes or generic action metric used previously.

The registered scientific claim did not pass. ATP’s hysteretic state exceeded the trajectory countermodel on utility and improved several endpoints, but it missed the action margin, used recurrence too often, lost decisively to the simpler final expected-value gate, failed to beat fixed forward and two other broad comparators at the registered margins, and encountered one failed recoverability condition. The current evidence therefore favors direct conditional expected value over hysteretic persistence for this task.

The next honest gate is not another adjustment to seeds 5018-5027. It is an external measured workload or independently authored benchmark in which the same trial stream supplies real action consequences, recurrence value varies within surface conditions, and a temporal-state controller faces modern sequence-based expected-value policies under nested selection. Final seeds 2001-2020 and both registered distribution shifts remain sealed.

20. The new problem: prediction is not intervention

Suppose a robot is assembling an object. Its force and torque stream may already support a rapid decision, or later measurements may clarify whether the assembly is failing. A model can be rewarded for predicting the final result from either window. That is useful, but it is not yet the same as showing that waiting, returning, retrying, stopping, or inspecting improves the physical result.

The missing operation is simple to state in plain language. The system must choose a processing route; that route must change an action; the action must actually occur; and its consequence must be measured. Only then can the route be evaluated as a control policy rather than a classifier.

This distinction matters directly to ATP. ATP is not merely the proposition that a longer sequence sometimes predicts better than a shorter sequence. Its stronger engineering form concerns the allocation of rapid transformation, structured return, correction, persistence, receiver preparation, action consequence, and cost. A dataset that observes only the original action cannot reveal what an unexecuted alternative would have done.

21. Three external routes and what each contributes

21.1 Physical wheel assembly

Macas, Trna, and Hanzlik independently recorded 524 sequences while a delta robot placed tires on small wheel rims [43]. Each sequence contains three force and three torque channels sampled at 0.004-second intervals and ends with a label indicating whether the finished assembly was acceptable. Provider metadata states that days 5 and 6 have human-generated labels, days 1 and 2 use labels

generated by a convolutional classifier that may be inaccurate, and days 3 and 4 have invalid labels. External-data preflight therefore excludes days 1-4. The eligible surface contains 260 human-labeled recordings: 60 on day 5 and 200 on day 6. The recordings were made in physical batches of four, yielding fifteen development groups and fifty external-evaluation groups. This is a real measured sequence problem with a real product outcome.

It is not yet an action-consequence experiment. The data do not record a randomized decision to continue, slow, recheck, retry, stop, or inspect. Assigning a cost to a false alarm or missed failure would create an operational model, not recover a measured consequence.

21.2 Handle-grasping failure diagnosis

The Toyota HSR archive contributes a different piece [44]. Its bounded raw record contains sensor and action traces, 100 ground-truth diagnosis documents, and 200 failure-event documents. The associated research asks how a robot can diagnose failed parameterizations and generate corrective experiences [45].

That is closely aligned with ATP’s correction question, but the evidence boundary remains. The bounded archive does not supply a trial-level table connecting an ATP route assignment, an actually executed correction, and the observed outcome after that correction. Paper-level reports cannot be attached to unrelated rows as if they were per-trial outcomes.

21.3 Robot-human handover

The handover corpus is the strongest declared expansion route [46]. It provides multimodal recordings, robot-action classes, human-action classes, and overall outcomes for 589 robot-to-human and human-to-robot trials. The provider already separates training, validation, and test labels, and the outcomes distinguish success from failures such as absent approach, absent grasp or release, and object drop.

The bounded intake preserved its provider metadata, datasheet, class names, wrench statistics, and all outcome labels. The raw episode archives were not downloaded. Even after download, the episodes remain observational with respect to ATP allocation unless route choice has a known assignment probability or an experimental intervention is added.

22. Registered measured-prediction pilot

22.1 A narrow question the public data can answer

The wheel data can answer: when predicting an observed assembly outcome, does a delayed causal sequence provide useful information beyond a fast causal prefix, and can a gate identify those cases without overusing delay?

The fast route ends at 25% of the normalized sequence. The delayed route ends at 75%. Neither route can use samples after its checkpoint. Calling the second route *delayed sequence processing* avoids the false implication that the source robot implemented biological recurrence.

22.2 Fully nested selection

Day 5 is the entire development surface. Its fifteen physical batches form five outer group folds. Inner group folds select sequence length normalization, causal receptive field, hidden width, regularization, optimization, and stopping. Only predictions made for batches excluded from route-model fitting may train the gate. Gate architecture and thresholds are selected by grouped cross-validation on those cross-fitted predictions.

After every choice is frozen, the selected routes and gate are refit on all day-5 batches and evaluated once on day 6. No day-6 outcome can change preprocessing, architecture, parameter count, threshold, endpoint, or interpretation.

Three sequence families are required because Development Study 4 did not exhaust modern temporal countermodels:

1. a causal temporal convolutional network [47];
2. a gated recurrent unit [48];
3. a causal self-attention encoder [49].

Their parameter counts and training budgets must be capacity matched within a frozen tolerance. The strongest selected direct expected-value gate and strongest selected trajectory expected-value gate are primary comparators. Fixed fast, fixed delayed, confidence, rate-matched random, and compute-matched policies remain visible.

Balanced accuracy, positive- and negative-class recall, Brier score, calibration error, route-use fraction, measured inference latency, and operation count are reported separately. A composite cannot turn a failed safety, calibration, or resource endpoint into a pass.

22.3 What a favorable pilot would establish

A favorable result would establish that conditional delayed processing improves prediction on an independently measured physical sequence corpus under one cross-day evaluation. It would not establish that ATP improves robot action, that recurrence causes the effect, or that the same allocation rule applies to a brain.

This pilot remains useful because it can reject weak gates before the more expensive physical intervention. It is a screening application, not the External-data preflight scientific action endpoint.

23. Prospective action-consequence application

23.1 The action must happen

At an eligible physical checkpoint, a safe experimental controller assigns a fast route or an extended sensing-and-return route. The route produces an actual supervisory or motor action: continuation, slowed execution, recheck, retry, abort, or inspection. The study records the complete observation stream, assignment probability, executed action, final task result, safety stop or damage, rework time, wall-clock latency, energy, and operator intervention.

Safe micro-randomization is preferred because route value is otherwise confounded by difficulty: a

controller naturally requests more processing on harder trials. If randomization is not permissible at some states, assignment propensities and safety exclusions must be logged before the outcome occurs.

23.2 Causal value is not an observed per-trial difference

Let $(Z_i=1)$ denote assignment to the extended route and $(Z_i=0)$ assignment to the fast route. Let $(Y_i(1))$ and $(Y_i(0))$ be the corresponding potential task losses. Only one is observed for each physical episode. Conditional route value is therefore

$$\tau(x) = \mathbb{E}[Y_i(0) - Y_i(1) \mid X_i = x]. \quad (37)$$

Positive $\tau(x)$ means the extended route reduces loss for states like x . The expression does not license describing an individual counterfactual difference as measured.

For a binary route with known or estimated propensity $e(X_i)$, a cross-fitted augmented inverse-propensity score for the mean outcome under route z is

$$\hat{\mu}_z = \frac{1}{n} \sum_{i=1}^n \left[\hat{m}_z(X_i) + \frac{\mathbf{1}[Z_i = z]}{\hat{e}_z(X_i)} \{Y_i - \hat{m}_z(X_i)\} \right], \quad (38)$$

where nuisance models are trained outside the scored fold [50]. Sequential decisions additionally require a trajectory-aware off-policy estimator rather than treating checkpoints as independent rows [51].

23.3 Heterogeneity must exist inside conditions

ATP requires more than one condition that always benefits from extended processing and another that never does. Within each eligible surface condition, the lower confidence bound for the fraction of positive conditional effects and the fraction of nonpositive effects must each exceed 0.10. Condition names cannot be supplied to the gate.

This replaces the synthetic fixture’s access to both route outputs with an honest causal statement. The study estimates conditional effects from randomized or propensity-logged episodes. It does not claim to observe both possible futures of one physical execution.

23.4 Consequences stay separate

Primary task success, damage, safety stop, rework duration, latency, energy, and operator intervention remain distinct. A composite may be added only when conversions are based on recorded operational costs frozen before evaluation. The primary paper cannot hide a safety loss behind faster average execution or an energy loss behind better calibration.

24. Power and the conjunctive gate

The application begins with a blinded feasibility pilot. Batch size, base event rate, within-batch correlation, and outcome variance determine the required sample. The plan targets at least 90% power for the smallest operationally meaningful task-loss difference after inflation for batch clustering and planned multiplicity.

ATP must then:

1. beat the strongest nested sequence-based expected-value policy on primary task loss beyond a frozen multiplicity-adjusted confidence margin;
2. remain noninferior on calibration, safety, latency, energy, and rework under their frozen margins;
3. avoid excess extended-route use beyond its frozen recurrence margin;
4. pass the within-condition heterogeneity rule;
5. preserve every adverse and null outcome.

A failure in any component is a failed conjunctive endpoint. A favorable predictive pilot cannot substitute for the missing intervention.

25. Prior failures remain active

Development Study 2 showed that a learned conditional-value gate could exactly reproduce the hand-specified ATP policy and that additional recurrence could sharpen probability without changing categorical action. Development Study 3 showed that learned routes and lower compute did not establish ATP-specific action superiority. Development Study 4 showed that nonlinear routes, generated task consequences, and temporal features still did not make hysteretic persistence superior to a simpler final expected-value gate.

External-data preflight does not erase those constraints. It responds by moving the question to independent measured data and by demanding an executed consequence. If the future experiment again favors direct expected value, that result must remain central.

26. Source-faithful and Apex interpretation

The historical ATP proposition remains a multidimensional question about rapid transformation, return influence, loss tolerance, ordering, correction, persistence, receiver preparation, action consequence, and resource constraint. Treating the Development Study 2-Development Study 4 controller failures as a complete verdict on that source would compress the proposition into a strawman. Treating public robot sequences as proof of ATP would compress the evidence in the opposite direction.

The strongest aligned interpretation preserves both truths. The source question is broader than a hysteretic gate, and every concrete gate must still answer to its failed tests. External data make the test more credible only when the task, action, and consequence are joined within one governed experiment.

27. External-data preflight disposition

The external-source preflight passed twelve structural checks. It verified source hashes, 527,799 wheel-data rows, 524 wheel recordings, 260 eligible human-labeled sequences, 100 handle-grasp diagnosis documents, 200 failure events, and 589 handover labels. It fit zero models and produced zero ATP, comparator, or recurrence-value outcomes.

The completed External-data preflight gate is therefore methodological rather than evidentiary:

PROTOCOL_AND_PUBLIC_DATA_ROUTE_READY / SCIENTIFIC_ACTION_GATE_NOT_RUN

This is the honest stopping point. The measured-prediction pilot and physical action-consequence application are now specified strongly enough to run without inventing an effect. The final benchmark and both distribution shifts remain sealed.

28. The problem before formal terminology

A controller can be described informally as switching between a rapid route and an extended sensing-and-return route. That description leaves several questions unresolved. Can the extended route be selected when no return evidence exists? What happens when the receiver is not prepared to use the returned information? Does the controller chatter when its value estimate lies between entry and exit thresholds? Is an unsafe trial silently assigned anyway? Does a recorded outcome count as a route intervention if the route never changed the action? Can better prediction be mistaken for better registered utility after latency and energy costs are included?

These questions are small enough to answer exactly. The formal certificate earns its terminology only after the operations are visible: receive a proposed route value, check whether return information exists, check whether the receiver can use it, apply separated entry and exit thresholds, refuse ineligible trials, and preserve the complete evidence history. Lean then checks the finite consequences of those definitions. It does not decide whether the definitions describe a brain or whether the proposed controller improves a physical robot.

29. A finite switching kernel

Let $(g_{t-1} \in \{F, R\})$ denote the previous fast or extended-return mode. Let $(A_t \in \{0, 1\})$ indicate that return information is available, $(B_t \in \{0, 1\})$ indicate that the receiver is ready, and let $(v_t \in \mathbb{N})$ be a discretized net route value after task benefit and declared cost have been combined. The certificate fixes $(h_{\text{off}} = 1)$ and $(h_{\text{on}} = 3)$, then defines

$$g_t = \begin{cases} F, & A_t = 0 \text{ or } B_t = 0, \\ R, & g_{t-1} = F \text{ and } v_t \geq 3, \\ F, & g_{t-1} = R \text{ and } v_t \leq 1, \\ g_{t-1}, & \text{otherwise.} \end{cases} \quad (39)$$

The checked consequences are exact. Extended output requires both return availability and receiver readiness. A fast state enters at the upper threshold but not at $(v_t = 2)$. An extended state exits

at the lower threshold but persists at ($v_t=2$). For every nonnegative number of repeated updates, constant eligible evidence at ($v_t=2$) preserves the starting mode. This is the bounded no-chatter result.

The certificate also defines a direct value route that selects extended processing whenever the same availability and readiness conditions hold and ($v_t \geq 2$), without carrying the previous mode. At ($v_t=2$), the direct policy selects extended processing while a previously fast hysteretic policy remains fast. Both satisfy the proved availability and receiver-readiness conditions. This supplies a machine-checked nonidentification result: the shared safety invariants do not select a unique controller and do not prove that hysteresis is required.

30. Confidence and utility are not route value

Development Study 2's equal-confidence construction is represented by two contexts with the same frozen confidence code, 125, but recurrence values 4 and 0. The former enters the extended route and the latter remains fast. The theorem does not claim that these values occur in a nervous system; it proves the logical point needed by the experiment: confidence alone cannot determine the route when two equal-confidence states can carry different expected consequences.

The cost boundary is checked with a second finite witness. One outcome has quality 90 and route cost 25; another has quality 80 and cost 0. Although the first outcome has higher quality, its registered utility is 65 rather than 80. Thus a sharper probability or better isolated prediction does not logically force better quality-cost performance. This mirrors the retained Development Study 2 observation that added recurrence sharpened target probability without improving the registered utility.

31. What an action-consequence record must contain

For episode (i), define six binary custody facts: a measured sequence (M_i), a recorded route assignment (Z_i), evidence that assignment changed the action (D_i), an executed action (X_i), an observed consequence (Y_i), and a known or governed assignment probability (P_i). The declared evidence-completeness predicate is

$$C_i = M_i \wedge Z_i \wedge D_i \wedge X_i \wedge Y_i \wedge P_i. \quad (40)$$

The formal profiles reproduce External-data preflight's source audit. The wheel-assembly corpus has measured sequences, executed original actions, and outcomes, but no ATP route assignment, route-caused action change, or assignment probability. The handle-diagnosis and handover profiles fail the same declared chain for their bounded ATP use. The prospective micro-randomized profile satisfies all six fields by construction. This proves only a schema-level distinction; it does not assert that the prospective experiment has occurred.

Potential outcomes remain separate from the observed episode. For fast and extended losses ($Y_i(0)$ and ($Y_i(1)$), the assigned route reveals only one coordinate:

$$O_i = \begin{cases} (Y_i(0), \emptyset), & Z_i = 0, \\ (\emptyset, Y_i(1)), & Z_i = 1. \end{cases} \quad (41)$$

Lean checks that one coordinate is absent for either assignment. This is why a trial-level causal difference cannot be described as directly observed, even when the final outcome is measured perfectly.

32. Refusal and endpoint separation

The prospective application must be able to decline an assignment. If safe micro-randomization is unavailable, or if the assignment is not logged before the outcome, the formal controller returns **refuse** rather than silently inserting the episode into the causal comparison.

Task loss is also not a sufficient representation of consequence. The declared vector is

$$\mathbf{c}_i = (L_i, D_i, S_i, W_i, T_i, E_i, H_i), \quad (42)$$

where the components represent task loss, damage, safety stops, rework, latency, energy, and operator intervention. Two checked fixtures have the same task loss but different damage, safety-stop, and intervention components; Lean proves that their vectors are unequal. The theorem prevents a task-score tie from erasing a safety difference.

The future scientific endpoint remains conjunctive:

$$G = G_{\text{task}} \wedge G_{\text{cal}} \wedge G_{\text{safety}} \wedge G_{\text{latency}} \wedge G_{\text{energy}} \wedge G_{\text{rework}} \wedge G_{\text{route}} \wedge G_{\text{heterogeneity}} \wedge G_{\text{history}}. \quad (43)$$

The all-true fixture passes. Changing only the safety component to false makes the complete gate false. A favorable average task result therefore cannot compensate for a failed safety margin inside the registered decision rule.

33. Losslessness and formal scope

The evidence-history object retains one Development Study 2 failed check, two Development Study 2 null results, four Development Study 3 failed checks, seven Development Study 4 failed checks, and zero External-data preflight action outcomes. Attaching the formal certificate is defined as the identity operation on that history, and Lean checks both identity and the exact counts. This makes adverse-result retention part of the formal interface rather than a prose promise.

The final benchmark remains **SEALED_NOT_OPENED**. A separate authorization function returns true only for the finite switching kernel and false for robot efficacy, biological recurrence, a SAN phase mechanism, and a physical action-consequence result. The formal package therefore strengthens the paper by narrowing what can be asserted, while preserving the broader historical ATP research question for the experiment that can actually test it.

34. Machine-check and proof boundary

The accepted source is `ATPInvariants.lean`, checked by the shared Lean compiler lease under Lean 4.30.0 [52]. The accepted run reports all thirty named theorem declarations, exits with code zero, and reports no `sorryAx`. Eighteen declarations report no axiom dependencies. Twelve rely only on the standard `propext` principle used by Lean when simplification rewrites propositions. The source, complete compiler output, proof-status ledger, claim signature, compatibility edges, and hashes are preserved together.

The formal leaf proves definitional and finite logical consequences of its explicit model. It does not verify the successive Python reference implementations, estimate an empirical effect, establish exchangeability or positivity, validate the public datasets beyond the bounded source audit, or prove that neural dynamics instantiate the gate. A future physical application must still produce the governed episode chain in Equation (40), estimate the route effect under the registered design, retain the strongest direct expected-value and modern sequence countermodels, and pass every component in Equation (43).

35. Formal kernel disposition

Formal kernel passes its bounded formal-verification gate. It makes the switching rules, refusal boundary, evidence-completeness requirement, counterfactual limitation, endpoint separation, adverse-history preservation, and authorization scope machine checkable. Its most important negative result is equally formal: availability and receiver readiness do not distinguish the hysteretic ATP controller from a direct expected-value policy.

The scientific action gate remains unrun. No final seed, distribution shift, or physical ATP action-consequence endpoint was opened. Strongest-form review should perform the complete source-faithful, strongest-form, nonduplication, and release-readiness audit; the release manuscript may standardize the manuscript and reproducibility package only if it preserves this empirical boundary rather than presenting the formal certificate as an efficacy result.

36. The problem before the comparison vocabulary

Two systems can look alike because both spend more computation on difficult inputs. They can nevertheless solve different problems. One system may repeat the same transformation until a halting probability crosses a threshold. A second may choose an early classifier. A third may compare a descending prediction with ascending evidence. A fourth may reopen a sensory-action loop because the receiving circuit is prepared to use returned information and the expected action improvement exceeds delay, energy, and safety costs.

Conversely, two systems can use different internal mechanisms while making the same decisions on every tested episode. In that case a new label does not yet identify a new causal operation. The distinction must be earned by an intervention, observable, or failure pattern that separates the candidate from its strongest neighbor.

This paper therefore asks four questions in order:

1. What complete problem did the dated ATP sources formulate?

2. Which parts are already owned by established computational and neuroscientific work?
3. What joined operation remains available to ATP as a distinctive hypothesis?
4. What result would discriminate that operation from a direct expected-value or adaptive-halting policy?

Only after those questions are answered is nonduplication terminology useful.

37. Strongest-form reconstruction of the ATP proposition

The dated record yields a ten-part operation rather than a networking slogan.

1. A receiver is already in a learned physical and computational state before an event arrives.
2. A rapid route may preserve usable continuity despite missing or corrupted events.
3. A return route can carry prediction, context, correction, gain, or persistence rather than a duplicate message.
4. Order and delay may change the meaning or usefulness of an event.
5. Repeated exchange can stabilize a state, but repetition can also waste time or reinforce an error.
6. The receiver’s readiness determines whether returned information can have the intended consequence.
7. Route selection must depend on expected downstream benefit, not confidence alone.
8. Latency, computation, metabolism, damage, and safety remain distinct costs.
9. A useful controller must sometimes refuse an unsafe or unlogged intervention.
10. The scientific endpoint is the consequence of an executed route-dependent action, not a retrospective prediction score alone.

This reconstruction passes the strawman-compression check because it preserves the 2017 traffic analogy, the 2019 neuronal correction, the earlier receiver-dependent sense–action ancestry, and the later PWD/NAPOT clarification without backdating later names. A controller that only chooses an iteration count instantiates part of this operation. A predictive-coding circuit that exchanges predictions and residuals instantiates another part. ATP denotes the proposed policy over the joined operation.

38. Four-lens comparison with the strongest neighboring work

38.1 Chronology and genealogy

Adaptive computation is not a later restatement of ATP. Graves introduced Adaptive Computation Time in 2016, before the April 2017 named ATP episode, and showed how a recurrent model could learn input-dependent numbers of internal updates [53]. The ATP genealogy instead begins with Micah Blumberg’s 2011–2012 receiver-dependent sense–action and neural-route sources, reaches the named Internet-transport analogy in 2017, receives an explicit neuronal correction in 2019, and is joined to PWD/NAPOT receiver mechanics in 2022. The chronologies overlap historically but arise through different source paths.

Predictive coding and reentrant-processing research precede both modern adaptive-computation

systems and named ATP [1,2]. ATP cannot claim priority for feedback, recurrence, prediction error, dynamic depth, confidence gating, early exit, or computation cost. Its priority question, if pursued, concerns the specific joined regime-allocation formulation evidenced by its dated sources. That is a document-relative genealogy claim requiring exact source comparison, not a universal firstness claim.

38.2 Narrative architecture

Adaptive Computation Time asks how many internal updates a recurrent model should perform before emitting an output [53]. PonderNet learns a probability distribution over halting steps while trading prediction against computation [57]. Early-exit networks learn when a shallower prediction is sufficient [54,59]. Universal Transformers recurrently refine token representations and may halt at different depths [55]. Learning-to-stop systems jointly learn predictions and a stopping policy [56]. Adaptive recurrent vision varies recurrent iterations with problem difficulty [58].

ATP’s narrative architecture begins one level earlier and ends one level later. It begins with a sender–receiver route whose return information may be absent, disordered, corrected, or unusable at the receiver. It ends with a route-dependent action and separately measured consequences. Computation depth is one coordinate. Direction, receiver readiness, controlled loss, correction content, persistence, safety refusal, and action consequence are additional coordinates.

Predictive coding supplies a different neighboring architecture: descending predictions alter lower-level expectations while ascending residuals revise higher-level states [1]. That architecture can operate with a fixed number of iterations and without an explicit policy over computation cost. ATP can use prediction error or surprisal as an allocation input, but it does not replace the content-specific predictive-coding account.

38.3 Mechanism and mathematics

The adaptive-computation literature generally defines a halting or exit variable over repeated transformations. ATP’s proposed state is a regime vector over heterogeneous operations. For episode (i), let

$$\mathbf{r}_i = (f_i, b_i, \ell_i, o_i, c_i, p_i, q_i, a_i, e_i, t_i), \quad (44)$$

where the coordinates denote forward influence, return influence, tolerated loss, order sensitivity, correction, persistence, receiver readiness, action coupling, energy, and latency. The symbols are measurement slots, not claims that unlike biological variables share a unit. A policy (π) maps the predecision history and receiver state into a feasible regime:

$$\pi : \mathcal{H}_i \times \mathcal{X}_i \longrightarrow \mathcal{R}_i^{\text{safe}}. \quad (45)$$

A binary fast/return controller is one projection of this larger object. ACT, early exit, and PonderNet primarily parameterize computation or stopping within a model. Predictive coding specifies the content relation between prediction and residual. ATP is distinct only if one or more of the additional regime coordinates have independent, measurable consequences after the strongest stopping, confidence, and direct-value policies are held fixed.

38.4 Evidence and consequence

The neighboring machine-learning papers report prediction, accuracy, generalization, and computation tradeoffs on their declared tasks [53–59]. The neural literature reports temporal, laminar, spectral, perturbational, and behavioral signatures of forward and return processing [1–13]. Those are positive evidence for the component mechanisms.

ATP’s completed evidence is mixed. Structured return content and receiver features affected several synthetic outcomes, but persistent hysteresis did not earn superiority over a direct expected-value policy. The physical action-consequence endpoint has not been run. This prevents an isomorphism claim between ATP and the neighboring systems and prevents an ATP-specific efficacy claim. The current comparison establishes a well-specified residual research question and a falsifiable route to answering it.

39. Atomic equivalence and the discriminator ATP must earn

A paper-level resemblance is too coarse for equivalence. Let the intervention signature of policy (π) across the registered test family be

$$\mathbf{s}(\pi) = (\hat{L}, \hat{C}, \hat{S}, \hat{T}, \hat{E}, \hat{R}, \hat{H}, \hat{V}), \quad (46)$$

where the components represent task loss, calibration, safety, latency, energy, route use, within-condition heterogeneity, and adverse-history preservation. Two policies are operationally equivalent for the present study only if their paired differences remain inside every preregistered equivalence margin under every required intervention and shift. Equality of route labels or average accuracy is insufficient.

The ATP-specific effectiveness claim requires a residual benefit beyond the strongest admissible comparator:

$$\Delta_{\text{ATP}} = \mathbb{E}[U\{\pi_{\text{ATP}}(H, X)\} - U\{\pi_{\text{best comparator}}(H, X)\}], \quad (47)$$

with the lower confidence bound exceeding the frozen margin while every safety, calibration, energy, latency, route-use, and heterogeneity condition also passes. The comparator family must include direct expected-value gating, adaptive halting, early exit, fixed and random allocation, and modern temporal models. If a simpler comparator matches or exceeds the complete signature, the ATP-specific operator is not identified on that domain.

This criterion protects the paper from two opposite errors. It does not discard the full ATP research question merely because one hysteretic implementation fails. It also does not count component-level recurrence or conditional computation as evidence that ATP’s joined operator is necessary.

40. Relationship to neighboring SAN programs

ATP is adjacent to several SAN programs, but each owns a different explanatory burden.

- **PWD calculus** owns the receiver-relative event difference and its candidate local measurements. ATP asks whether that difference justifies a different communication regime.
- **From Gamma Synchrony to Informative Difference** owns conditional information relative to tonic expectation. Surprisal can enter ATP’s value estimate without determining the route by itself.
- **NAPOT methods** own multi-array reconstruction and observability. ATP asks how much rapid or return processing a reconstruction recruits.
- **The Flow of Information in the Brain** owns the laminar, corticothalamic, long-range, motor, and sensory-return traffic architecture. ATP is a policy over traffic conditions, not the anatomy itself.
- **The Coadaptive Substrate Principle** owns neural drift and the read/write interface’s mutual adaptation. ATP asks which communication regime is selected during a bounded read or write operation.
- **Neural Tuning** owns learned receiver selectivity. Receiver tuning is an input to ATP rather than the switching policy.

The paper remains distinct only while these divisions produce separable measurements and perturbations. If ATP reduces to PWD magnitude, surprisal, NAPOT reconstruction quality, anatomical direction, or tuning strength, it should be merged rather than retained as an independent mechanism.

41. Strongest countermodels and retained adverse evidence

The direct expected-value policy is the present strongest countermodel because it uses the same route evidence and declared consequences without carrying hysteretic state. Development Study 2 showed exact agreement between a learned benefit gate and the hand-specified ATP policy. Development Study 4 showed that direct final expected value produced higher registered utility than the hysteretic implementation. Formal kernel proved that the two policy classes can obey the same availability and readiness constraints yet diverge in the mid-band. These results jointly show that receiver awareness and safety eligibility do not imply that persistence is useful.

Adaptive halting supplies a second strong countermodel. If input-dependent ponder time or an early-exit network reproduces the complete intervention signature, then variable compute—not ATP’s broader regime vector—may explain the result. Predictive coding supplies a third: if explicit prediction and error units explain the observed return effects without a separate allocation policy, the content-specific circuit should receive credit. Fixed recurrent and modern sequence models remain necessary because a conditional policy cannot claim value merely by comparing itself with a weak feedforward baseline.

Every observed failure remains live evidence:

- Development Study 1 overused recurrence and lost to a confidence policy.
- Development Study 2 tied the learned gate and failed cost neutrality.
- Development Study 3 failed four registered checks.
- Development Study 4 failed seven registered checks and favored direct final expected value.
- External-data preflight produced no physical action-policy outcome.
- Formal kernel established policy nonidentification under shared safety invariants.

The source proposition remains broader than these implementations, but the implementations remain the evidence for any effectiveness claim. Neither layer erases the other.

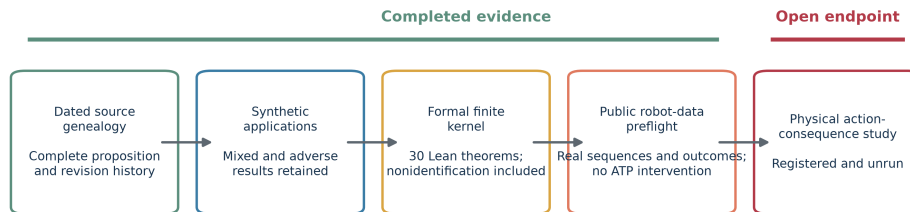
42. What the paper can claim now

The paper can claim that ATP supplies a dated, source-recovered framework for treating communication as receiver-dependent regime allocation rather than context-free signal transport. It can claim that the framework has been converted into measurable variables, synthetic tests, strong countermodels, a finite formal kernel, refusal rules, and a prospective physical action-consequence protocol. It can report that structured return and conditional allocation mattered in several synthetic fixtures while the tested hysteretic controllers failed to establish unique effectiveness over learned direct-value policies.

The paper cannot yet claim that a brain implements the finite controller, that ATP improves a physical robot, that a phase or oscillatory mechanism causes the route decision, or that hysteresis is necessary. Those questions are not weakened by being stated accurately; they are made testable.

43. Evidence boundary for the present preprint

Evidence classes and the remaining causal gap



A complete governed episode must contain: measured stream -> route assignment -> route-caused action change -> executed action -> observed consequence -> known assignment probability.

The preprint can establish the framework, negative results, formal boundaries, and experimental design without claiming that the final physical test has occurred.

The present evidence supports a bounded theory, methods, synthetic-application, formal-verification, and registered-experiment preprint only while the paper:

1. reconstructs the problem before naming ATP;
2. preserves the dated 2011–2022 genealogy without backdating later vocabulary;
3. gives adaptive computation, early exit, predictive coding, and recurrent processing their established contributions;
4. keeps the direct expected-value countermodel central;
5. reports every adverse, null, tie, failed gate, and unrun physical endpoint;
6. distinguishes mathematical invariants, synthetic application results, public-data preflight, and prospective physical evidence;

7. contains a complete claim-source matrix and four-lens comparison ledger;
8. provides inspectable architecture and evidence-flow figures;
9. ships the code, frozen contracts, data receipts, compiler logs, manifests, and hashes needed to reproduce the bounded results;
10. labels the physical action-consequence study as registered and unrun.

These conditions permit a scientifically useful preprint before the physical trial because the paper’s contribution includes the clarified framework, negative application record, formal boundaries, and executable experimental design. A later physical result would require a new version, not a silent replacement of this record.

44. Conclusion

The strongest-form and nonduplication review establishes the present evidentiary boundary. ATP is not treated as a synonym for recurrence, feedback, predictive coding, adaptive computation, early exit, or variable depth. Its remaining distinctive hypothesis is the joined receiver-dependent allocation of direction, controlled loss, ordering, correction, persistence, cost, and route-caused action consequence. That hypothesis is clearly separable in the registered design but is not yet established by a physical intervention.

ATP therefore remains a falsifiable proposal for joined receiver-dependent regime allocation, supported by a complete source genealogy, measurable variables, bounded formal invariants, and mixed executable evidence. The strongest tested controller did not outperform direct learned expected value, and the decisive physical action-consequence study remains registered and unrun. The preserved final seeds and shifts remain closed. A later physical result must appear as a new version that retains this negative and mixed record rather than silently replacing it. The current status is BOUNDED_PREPRINT_READY / SCIENTIFIC_ACTION_GATE_NOT_RUN.

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